3 Lesions That Mimic Endodontic Pathosis

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CHAPTER OUTLINE

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Summary

Importance of Differential Diagnosis

It has long been recognized that periapical cysts and granulomas make up the majority of radiographic lesions in the tooth-bearing regions of the jaws.[17,28,137,243] Many of these radiolucencies are asymptomatic and discovered on routine radiographs. For clinicians, it is easy to assume that any periapical radiolucency associated with pain is of pulpal or inflammatory origin, especially if the tooth has a significant restorative history. However, numerous disease entities of variable clinical significance can be identified in the periradicular regions, and differentiating the typical endodontic presentation from that of a potentially more life-threatening or insidious etiology is of critical importance to ensure ideal management and outcome (Fig. 3.1).[139,141,187,216,235]

Limits of Pulp Testing

The use of pulp testing in conjunction with medical history, dental history, clinical examination and radiographic findings is considered the standard routine to differentiate pulpal pathology from that of a noninflammatory or nondental etiology.[73] In spite of this, pulp testing has significant limitations as historical studies suggest there is not a correlation between the histologic status of the pulp and a patient's clinical signs and symptoms.[228,229] However, a more recent investigation reported a correlation between the histologic and clinical diagnosis of irreversible pulpitis in 84.4% of cases and agreement was even higher for normal pulps and reversible pulpitis.[209] It is not uncommon to encounter patients with extensive caries throughout the dentition, suggesting significant pulpal inflammation, but no history of pain (Fig. 3.2).

The accuracy of diagnostic testing is further complicated by a lack of proprioceptive fibers in the dental pulp, which results in the patient's inability to localize the source of pain.[167,228] Thermal and electric pulp testing are actually a measure of the pulp's ability to sense a stimulus rather than a representation of overall health, and these measurements are susceptible to both false-positive and false-negative responses.[5,98,118,119,168,197,264] Given the diagnostic limitations of pulp testing, many clinicians place an undue amount of importance on radiographic findings.

Limits of Radiology

According to a 2012 systematic review of cross-sectional studies evaluating the periapical status of over 300,000 teeth, 5% of all teeth had periapical radiolucencies.[188] The prevalence of periapical radiolucencies in the population illustrates

the critical importance of being able to formulate an accurate differential diagnosis of these radiographic abnormalities. Historically, radiographs could not demonstrate periapical lesions located in cancellous bone until mineral loss resulted in a lesion of 1 to 7 mm.[18,20] This suggests the prevalence of periapical inflammatory disease is even higher because all teeth with inflamed or necrotic pulps will not result in a periapical radiolucency. Different anatomical locations within the oral cavity have variable likelihoods for identification of a radiolucent lesion based on the thickness of the cortical plate and volume of bone.[19] Additionally, the actual size of a radiolucent defect is larger clinically than can be appreciated with 2D radiographic imaging.[230] In less common clinical situations, the periodontal ligament (PDL) space can be widened or displaced prior to pulp necrosis due to the cytokine cascade from an inflamed pulp.[2,270] Because of the technical limits of traditional imaging, a 2D radiograph should not be considered a panacea in the diagnostic process.



Fig. 3.1 A well-defined radiolucency near the apex of the maxillary first premolar and canine. Although the premolar did not respond to pulp testing, the location of the lesion and clinical presentation were suspicious and a biopsy revealed an odontogenic keratocyst.

The interpretation of 2D films and digital radiographs is subjective due to significant intraobserver and interobserver variability.[32,95,248] Three-dimensional cone-beam computed tomography (CBCT) technology is a more accurate way of identifying periapical inflammatory disease,[10,57,75,150,193,259] but the interpretation of 3D images is also subjective and based on the experience of the clinician and quality of the images.[16,190,202,254] Moreover, clinicians must be able to identify all of the incidental findings that may be discovered with these larger data sets.[67,185,210] The American Association of Endodontists and the American Academy of Oral and Maxillofacial Radiology published a Joint Position Statement in 2015 providing 14 scientifically based guidelines for the use of CBCT in endodontic treatment. This position paper and other reports recommended the judgment of the treating clinician should determine the need for a CBCT and suggested that patient history and clinical examination should demonstrate that the benefits of the scan outweigh the potential risks.[1,151] "CBCT should not be used routinely for endodontic diagnosis or for screening purposes in the absence of clinical signs and symptoms. Clinicians should use CBCT only when the need for imaging cannot be met by lower dose two-dimensional radiography."[1] Furthermore, when indications arise for the use of CBCT a limited field of view is recommended to minimize the amount of radiation and maximize the image resolution. A CBCT can serve as a valuable tool to differentiate subtle radiographic changes and facilitate the development of a more comprehensive differential diagnosis compared to conventional radiography (Fig. 3.3).[57,192]

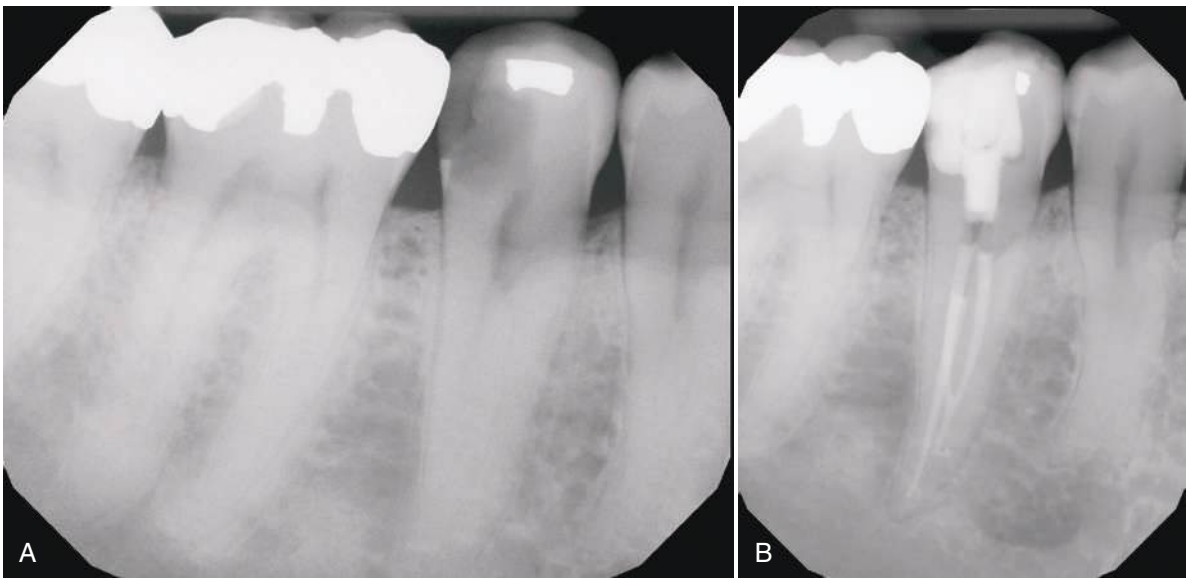


Fig. 3.2 A, Deep caries in the mandibular second premolar, but the patient was asymptomatic with no history of pain. **B,** Tooth #29 responded normally to both cold and electric pulp testing, but nonsurgical root canal treatment was completed for restorative reasons.

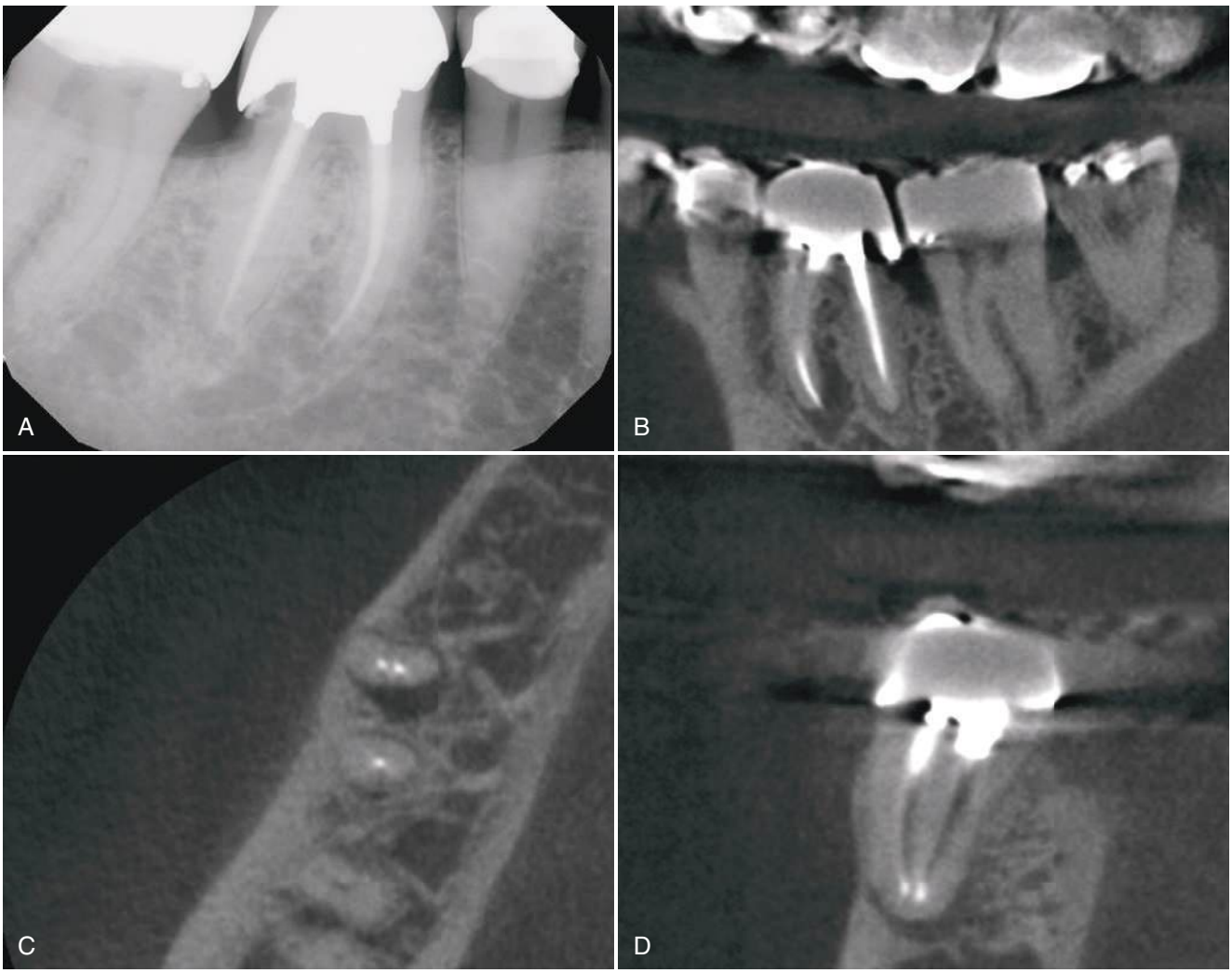


Fig. 3.3 A limited field CBCT is a useful adjunct when the findings on a periapical image are not conclusive. **A,** A subtle radiolucent area is present at the mesial root apex of the mandibular first molar. **B,** Sagittal, **(C)** axial and **(D)** coronal CBCT images reveal a more prominent loss of density at this root apex.

Differential Diagnosis of Periapical Pathosis

In many cases, the radiographic findings associated with noninflammatory pathologies can very closely mimic the presentation of the more common periapical cyst or periapical granuloma, and there is no way to definitively determine the clinical importance of a lesion based on radiographic features alone.[17,137,139,141,235,243] However, accurate interpretation of the bone abnormalities in conjunction with a thorough review of the medical history, dental history, pulp testing, and clinical examination can yield a differential diagnosis that could result in earlier treatment and potentially a better prognosis. Given the limitations of both pulp testing and radiographic interpretation, the diagnosis of periapical inflammatory disease is associated with a variable level of uncertainty. As a result, using a repeatable method when evaluating radiographic abnormalities can aid the clinician in developing a comprehensive differential diagnosis.

MULTILOCULAR RADIOLUCENCIES

Periapical cysts and periapical granulomas have a wide spectrum of radiographic presentations with some causing extensive bone destruction and others resulting in minimal radiographic abnormalities (Fig. 3.4). Irrespective of size, the presentation of periapical inflammatory disease should be radiographically unilocular and confirmed with a lack of responsiveness to pulp testing. If a multilocular radiolucency is identified on a periapical image, then a panoramic radiograph or CBCT is recommended to confirm the multilocular growth pattern and extent of the lesion. The differential diagnosis of multilocular radiolucencies is extensive, but the most common are odontogenic keratocyst (OKC), ameloblastoma, and central giant cell lesion (CGCL) (Box 3.1).

Odontogenic Keratocyst

The OKC is a developmental cyst that accounts for 3% to 11% of all odontogenic cysts.[42] In 2005, the World Health Organization reclassified the OKC as a neoplasm with the

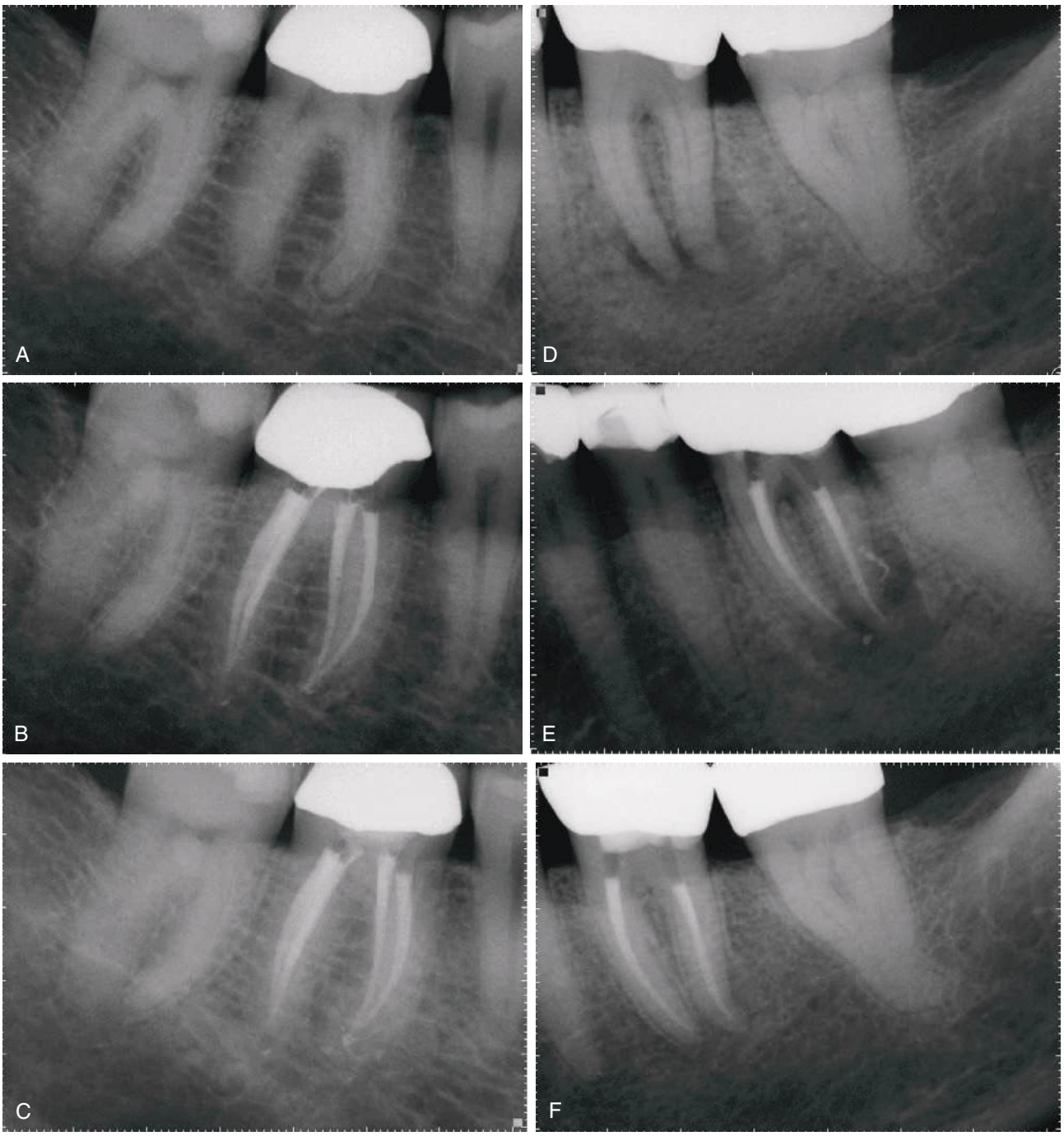


Fig. 3.4 Periapical inflammatory disease presents with a wide spectrum of radiographic changes. **A,** Preoperative periapical image of a mandibular first molar with a necrotic pulp and asymptomatic apical periodontitis. There is a widened PDL space and subtle radiolucency at the mesial root apex. **B,** Nonsurgical root canal treatment was completed. **C,** Evidence of bone healing at 6-month recall. **D,** Preoperative periapical image of a mandibular first molar with a necrotic pulp and asymptomatic apical periodontitis. This lesion is considerably larger than in image **A** and involves the mesial and distal root apices with coronal extension along the distal aspect. **E,** Nonsurgical root canal treatment was completed and a lateral canal is present in the apical half of the root. **F,** Evidence of bone healing at 6-month recall.

Box 3.1 **Uncommon Multilocular Radiolucencies**

Ameloblastic fibroma
Odontogenic myxoma
Central odontogenic fibroma
Calcifying epithelial odontogenic tumor
Orthokeratinized odontogenic cyst
Lateral periodontal cyst
Calcifying odontogenic cyst
Central hemangioma/Arteriovenous malformation
Aneurysmal bone cyst
Cherubism
Hyperparathyroidism
Intraosseous mucoepidermoid carcinoma

designation keratocystic odontogenic tumor because of molecular genetic changes found in other neoplasms, but in 2017 the name was changed back to the original OKC.[69,238] As a result, both names will appear in the literature and it is important to distinguish OKCs from lesions of inflammatory origin due to substantial growth potential and recurrence risk.

The OKC can be identified at any age, but most are discovered as a solitary lesion between the ages of 10 and 40.[42] If multiple OKCs are present or a single OKC is diagnosed before the age of 10, nevoid basal cell carcinoma syndrome would be suspected.[81] Histologically, the basal layer of the cyst lining has palisaded cuboidal cells, often with hyperchromatic nuclei, and the surface of the cyst exhibits parakeratin production. The OKC has a thin, noninflamed, stratified squamous epithelial cyst lining consisting of six to eight layers of cells that is easily dislodged from the underlying connective tissue during surgical treatment increasing the likelihood for recurrence. The growth pattern also influences the reported recurrence rates because a multilocular presentation is more difficult to completely remove with curettage.[121] Most reports indicate a recurrence rate of approximately 30%.[23,49] However, the rate of recurrence is variable, depending on the study protocol, and recurrence can occur more than 5 years after the original surgical intervention, suggesting long-term follow-up is mandatory.[23,81]

Radiographic Features. The OKC presents as a well-defined radiolucency and can be found at any location in either jaw (Fig. 3.5).[6,91,216] They can occur in the periapical region, but there is an uneven distribution of OKCs throughout the maxilla and mandible with the highest incidence in the posterior body and ramus of the mandible (Fig. 3.6).[182,250] When an OKC is associated with an impacted tooth (25% to 40% of cases), it should not be confused with a lesion of endodontic origin due to its location surrounding the crown. This presentation would be indistinguishable from that of a dentigerous cyst (Fig. 3.7). Smaller OKCs are most often asymptomatic and commonly present as a unilocular radiolucency while larger OKCs tend to have the classic multilocular appearance.[250] Larger OKCs may present with pain and swelling, but many are asymptomatic with no evidence of bone expansion.

Ameloblastoma

Ameloblastoma is the most common odontogenic tumor. It presents over a wide age range with an equal frequency after the age of 20, but it is uncommon before 20 years of age and rare before the age of 10.[122] These benign tumors are slow growing but can become large and disfiguring if left untreated. There are six histologic subtypes, but these differences have no impact on the treatment options or recurrence rate.[31] Ameloblastomas will often infiltrate normal bone resulting in extension beyond radiographic margins and higher recurrence rates. A more aggressive surgical resection with a larger margin will result in a lower recurrence rate, but this may not be possible due to the size of the tumor or its location near important anatomical structures. Marginal resection is the most common treatment, but recurrence rates of 15% have prompted some surgeons to recommend a 1 to 2 cm resection margin due to the infiltrative growth pattern. However, when the tumor presents near vital structures, a resection may not be possible and the resulting curettage is associated with a recurrence rate that ranges from 50% to 90%.[94] As the molecular pathways involved in the pathogenesis of ameloblastoma are better understood, it may be possible to block these mechanisms with targeted chemotherapeutics, preventing the need for surgery.[109,219] As with OKCs, recurrence can occur many years following the initial surgical intervention and long-term follow-up is mandatory.

Radiographic Features. Ameloblastoma can occur at any location in the maxilla or mandible, but 66% are found in the mandibular posterior region and only 15% present in the maxilla.[179] Smaller lesions are often asymptomatic and

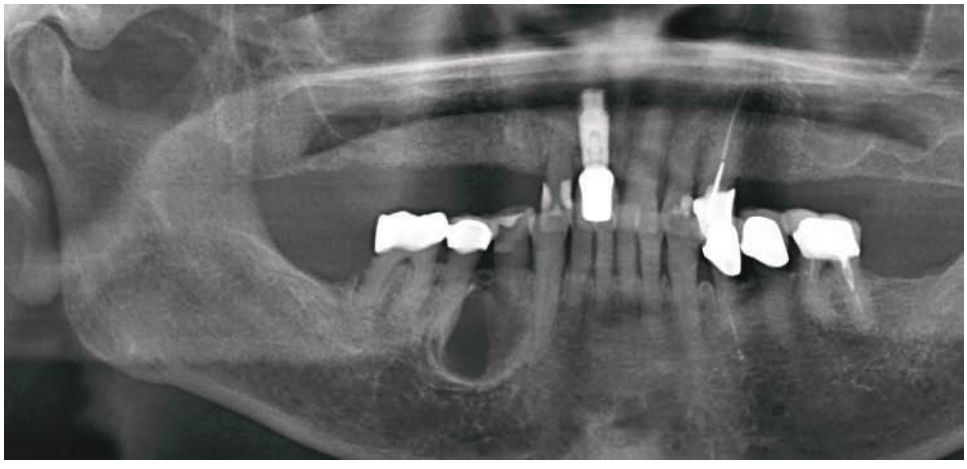


Fig. 3.5 Although most odontogenic keratocysts occur as multilocular radiolucencies at the angle of the mandible, they can be located at any position in either jaw. A cropped panoramic radiograph reveals a corticated odontogenic keratocyst in the mandibular right premolar region. (Courtesy Dr. Christel Haberland.)

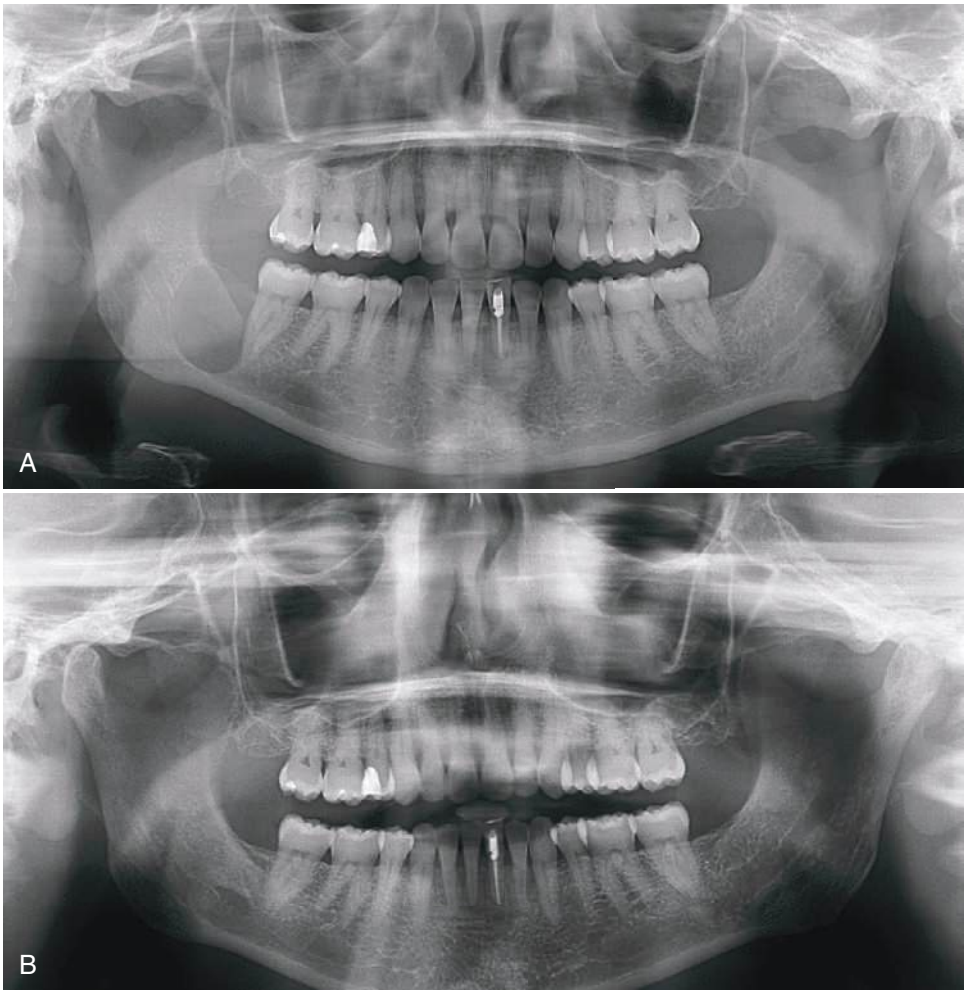


Fig. 3.6 A, A cropped panoramic radiograph shows a well-defined unilocular odontogenic keratocyst at the right angle of the mandible and distal root of second molar. The lesion was surgically removed and good bone healing was evident at a 3-year recall with no indication of recurrence **(B).** (Courtesy Dr. Christopher Bacsik.)

may present as a unilocular radiolucency found on routine radiographic examination. A small, unilocular ameloblastoma in the periapical region may be indistinguishable from a periapical cyst or granuloma.[77,97] Larger ameloblastomas often result in a painless swelling, and expansion of the cortical plate may help differentiate an ameloblastoma from both inflammatory lesions and OKCs (Fig. 3.8). As with OKCs, many ameloblastomas are associated with an impacted tooth and should not be confused with a lesion of endodontic origin. Ameloblastomas may cause resorption of adjacent teeth, and irregular scalloping of the margin is highly suggestive of the diagnosis.[48] However, when an ameloblastoma presents as a typical nonexpansile multilocular radiolucency, it cannot be distinguished from any other lesion with this growth potential based on radiographic findings alone.

Central Giant Cell Lesion

Long recognized by the term *central giant cell granuloma*, the most recent World Health Organization classification now designates this entity as CGCL.[69] Like OKCs and ameloblastomas, the CGCL will often present as a multilocular radiolucency. It has an unknown etiology that may be a reactive process, but a benign neoplastic origin has not been ruled out. The diagnosis of CGCL is made over a wide age range, with the majority discovered before the age of 30.[172,265] Most case series report a female predilection, and 70% of CGCLs are located in the mandible with a tendency to cross the midline (Fig. 3.9).[143] Histologically, the CGCL is identical to the brown tumor of hyperparathyroidism, and the distinguishing microscopic feature is the presence of multinucleated giant cells. The number and size of the giant cells vary considerably throughout the lesion, and there is controversy about the prognostic significance of this finding, with some studies suggesting a more aggressive behavior.[143,265] Curettage is the most common treatment for CGCL, but a more extensive surgery may be needed for aggressive or recurrent lesions. Alternative treatments for CGCL include, but are not limited to, intra-lesional corticosteroids, bisphosphonates, and imatinib.[54] The overall recurrence rate for CGCL is variable, with most studies reporting a rate near 20%.[53]

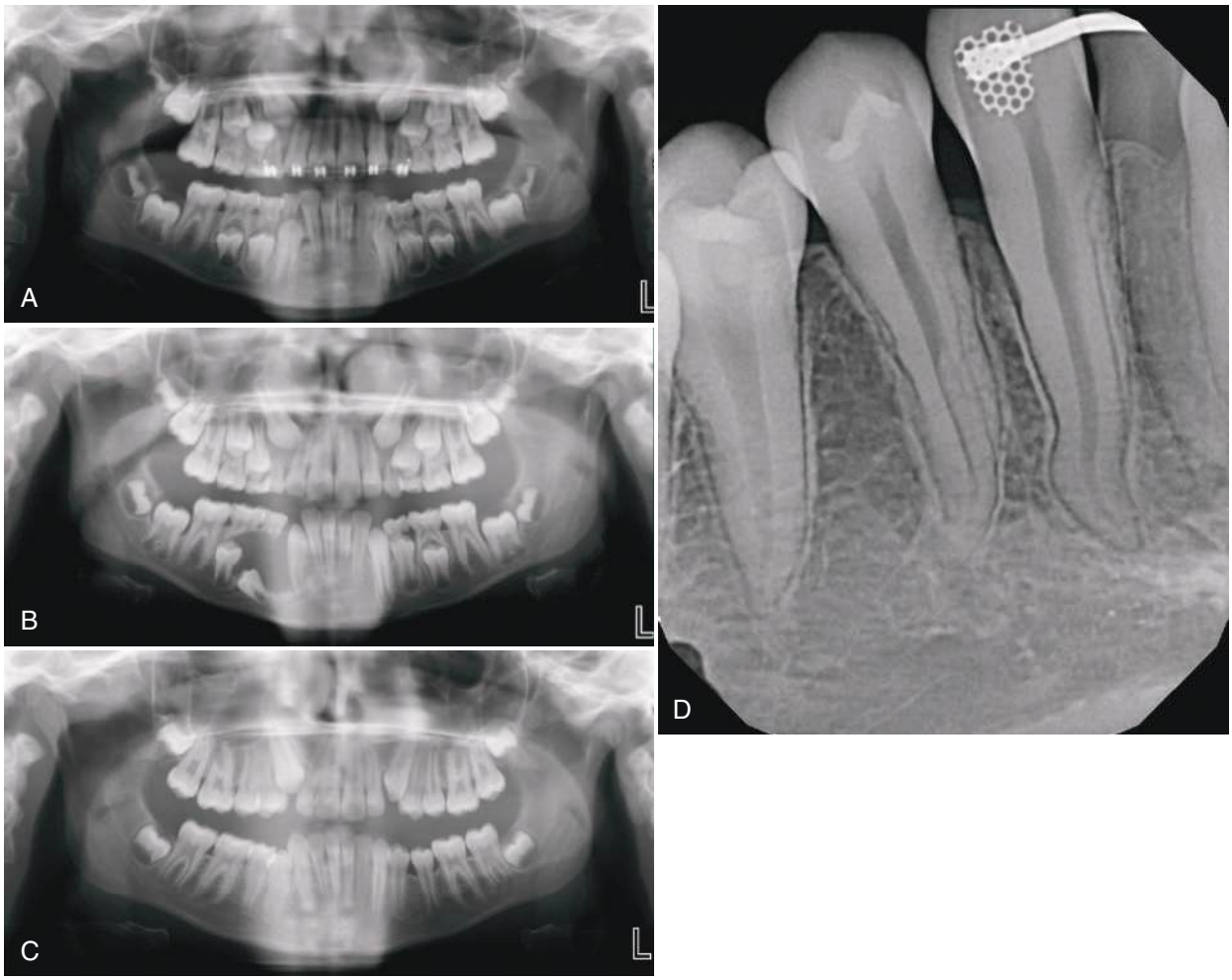


Fig. 3.7 **A,** A panoramic radiograph of a 10-year-old female with a mixed dentition. **B,** Serial panoramic images were obtained for orthodontic reasons and a well-defined expansile lesion developed in the mandibular right premolar region over 9 months. This type of radiolucency should not be confused with an inflammatory etiology due to its location around the crown of an unerupted tooth. This odontogenic keratocyst was treated by unroofing the cyst without removal of the entire lesion. **C,** Panoramic image with no evidence of recurrence at 2 years. **D,** Periapical image with no evidence of recurrence at 6 years.

Radiographic Features. The characteristic presentation of CGCLs can be subdivided into two categories.[44] Nonaggressive CGCLs tend to be asymptomatic and are often discovered on routine radiographs or due to a subtle expansion. These lesions are smaller and grow more slowly than the aggressive subtype. Additionally, nonaggressive CGCLs should not cause perforation of the cortical plate or resorption of roots. Aggressive CGCLs are more likely to exhibit pain and rapid growth.[53] Aggressive CGCLs may result in perforation of the cortical plate, resorption of roots, paresthesia, and extension into soft tissues following perforation. These lesions are usually larger at the time of diagnosis and exhibit a higher recurrence rate.

Most commonly, a CGCL will have a well-defined but noncorticated border. The size can vary from several millimeters to more than 10 cm. As a result, the amount of expansion and thinning of the cortical plate will be highly unpredictable as will the number and size of the loculations (Fig. 3.10). When a smaller lesion is discovered in the periapical region, it can look identical to a periapical cyst or periapical granuloma and larger CGCLs may resemble both OKC and ameloblastoma.[51,226]

RADIOPACITIES IN THE PERIAPICAL REGION

Although a multilocular radiolucency is consistent with a noninflammatory etiology, any of the multilocular radiolucencies can also present as a unilocular lesion in its earlier stages prior to reaching full growth potential. As a result, other radiographic features need to be considered when forming a differential diagnosis. Endodontic lesions of inflammatory origin most often present as radiolucent lesions with no indication of radiopacity.[21] As a result, if any evidence of radiopacity is identified, suspicion of a noninflammatory etiology should be elevated, especially if a limited restorative history is present or pulp testing is inconclusive. Exceptions to this rule would include condensing osteitis and maxillary sinusitis of endodontic origin (MSEO).

Condensing Osteitis

Condensing osteitis is an area of bone sclerosis in the periapical region that results from an inflammatory stimulus secondary to chronic pulpal inflammation or necrosis. It is most commonly diagnosed in children and young adults, but it can occur in older individuals. Compact bone replaces the fatty marrow, and there are variable levels of fibrosis

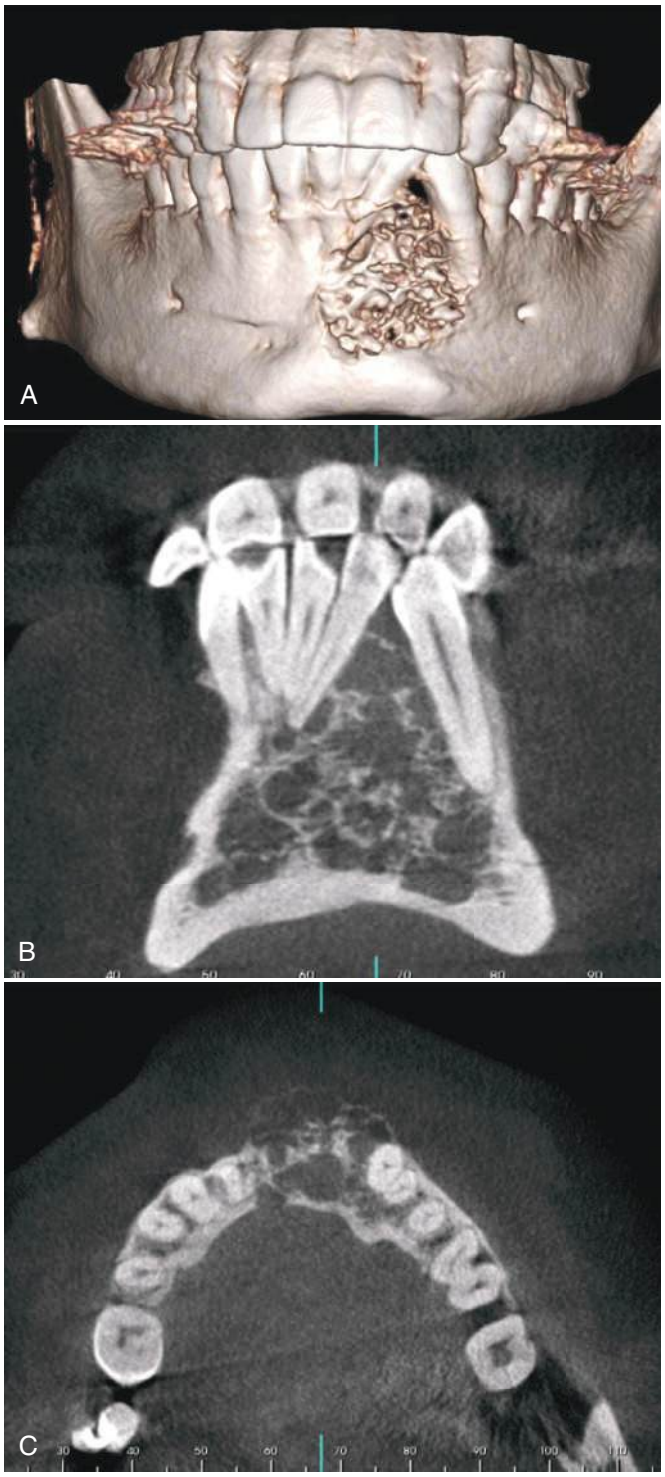


Fig. 3.8 A, 3D volumetric rendering of an ameloblastoma between roots of mandibular canine and mandibular lateral incisor with destruction of the buccal cortical plate. **B,** Sagittal CBCT image reveals a low-density lesion with numerous septae displacing the roots of teeth #22 and #23. **C,** Axial CBCT image exhibiting a perforation of the buccal cortical plate and buccal-lingual expansion that is common with ameloblastomas. This lesion would be characterized as a multilocular radiolucency on a periapical image. (Courtesy Dr. Bruno Azevedo.)

consistent with an osteoblastic response to a low-grade stimulus.[99] Treatment is focused on the removal of the inflammatory stimulus, and Eliasson et al. reported 85% of these lesions resolve to some extent following root canal treatment or extraction of the compromised tooth.[68] This

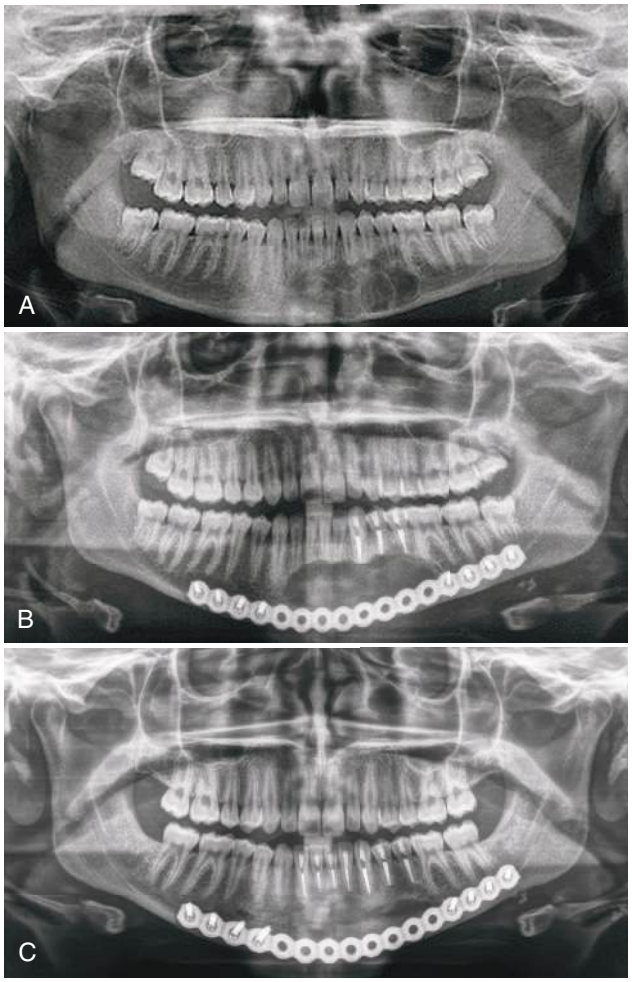


Fig. 3.9 A, Central giant cell lesion extending from left mandibular premolar region across the midline in a 17-year-old female. This radiolucency cannot be differentiated from other multilocular lesions without a biopsy (see Box 3.1). **B,** The left mandibular premolars and canine had nonsurgical endodontic treatment prior to removal of the central giant cell lesion and the mandibular incisors had nonsurgical endodontic treatment after removal of the lesion, when it was determined intraoperatively the blood supply to these teeth had been compromised. **C,** Eighteen-month recall with evidence of increased bone density, but incomplete healing.

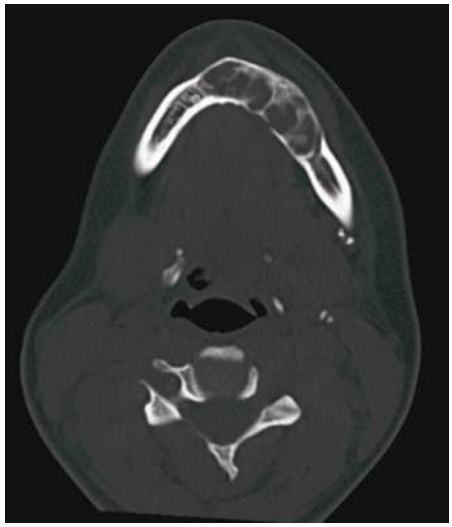


Fig. 3.10 Axial view of CT showing the extent of the central giant cell lesion of the same patient from Fig. 3.9. This plane reveals a multilocular growth pattern with thinning of both the buccal and lingual cortical plates.

study is limited by a small sample size, and it is should be noted that regression of the dense bone did not correlate with the quality of the root canal fillings.

Radiographic Features. The true incidence of condensing osteitis in the population is unknown, as there is a small amount of literature on the condition, and historically it has likely been confused with idiopathic osteosclerosis due to similar radiographic appearances and clinical presentation. Condensing osteitis most commonly presents in the premolar and molar regions of the mandible.[76] The involved tooth exhibits a widened PDL space and is associated with a localized radiodense zone apical to a carious or heavily restored tooth (Fig. 3.11). The sclerotic bone is asymptomatic, but the involved tooth may be painful due to pulpal inflammation or necrosis.

Maxillary Sinusitis of Endodontic Origin

MSEO is an inflammatory disease of microbial origin caused by a necrotic pulp or a root canal–treated tooth with posttreatment disease and may account for 10% of sinusitis cases.[169] The teeth most closely associated with the maxillary sinus include the second molar, first molar, and second premolar. Eberhardt et al. reported that the distance from the MB root of maxillary second molars to the sinus floor is less than 1 mm, while Tian et al. measured similar distances between root ends and the sinus floor but also found that this difference increases with age.[65,249] More than 40% of maxillary molars have a root that protrudes into the sinus.[125] Because of the close anatomical proximity of some root ends to the maxillary sinus and the thin plate of bone separating the roots from the sinus lining, an endodontic infection can cause bone destruction leading to a communication through the floor of the sinus. This communication can result in the accumulation of inflammatory mediators elevating the sinus lining or triggering an osteoblastic reaction. Other potential bacterial pathways as an etiology for sinusitis include periodontal disease, root fractures, extractions, and the placement of dental implants.

The sinus component is often asymptomatic and, although the pulpally involved tooth may be painful, the typical symptoms of an endodontic infection may not be obvious. Because MSEO results from a necrotic pulp or a tooth with posttreatment disease, there will be no thermal sensitivity. Swelling and percussion symptoms are less common due to the release of pressure into the sinus. Headache, facial pain, unilateral nasal obstruction, postnasal drip, and nasal discharge are clues to the presence of MSEO, especially when these symptoms are concurrent with a bacterial etiology of endodontic origin.[268,269]

In spite of a reported prevalence ranging from 1.5% to 14%, MSEO often goes undiagnosed, and the literature suggests dental infections contribute to a large percentage of maxillary sinusitis cases.[33,160,179,191] If MSEO is not diagnosed, patients may be erroneously prescribed multiple rounds of antibiotics or undergo unnecessary sinus surgeries due to chronic sinus infections. Treatment involves the elimination of the bacterial etiology with root canal treatment, endodontic surgery, or extraction.[183] Any attempt to directly treat the sinus with a surgical approach will be ineffective, as the source of the infection will not be directly addressed. Likewise, systemic antibiotic therapy may reduce symptoms temporarily, but will not result in a long-term cure.

Radiographic Features. MSEO is often difficult to identify with conventional imaging due to the superimposition of root ends, sinus, cortical plates, and the zygoma (Fig. 3.12). As a result, a high percentage of MSEO cases are missed with routine periapical radiographs.[184,246] A panoramic radiograph may exhibit a sessile, slightly radiopaque dome-shaped elevation. Limited field CBCT will detect a greater percentage of cases of MSEO than other imaging modalities, and two distinct patterns can be identified.[160]

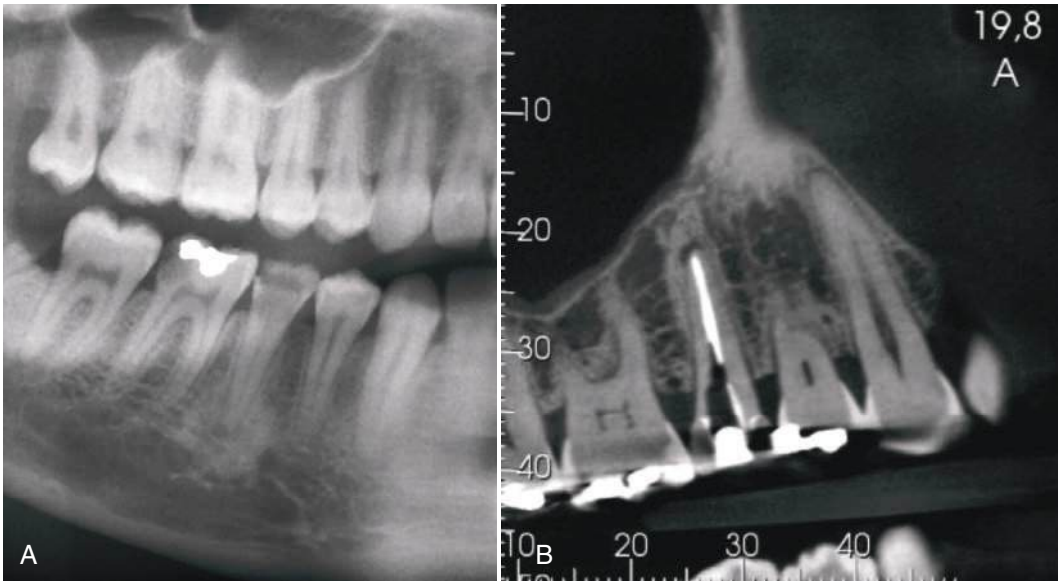


Fig. 3.11 A, Cropped panoramic image of condensing osteitis exhibiting a widened PDL space with a radiodense rim at the apex of a carious tooth #29. **B,** Sagittal CBCT image of maxillary premolar with previous endodontic treatment. A homogeneous high-density lesion is present with circumferential loss of density at the root apex suggesting an inflammatory etiology. (Courtesy A, Dr. Christel Haberland; B, Dr. Bruno Azevedo.)

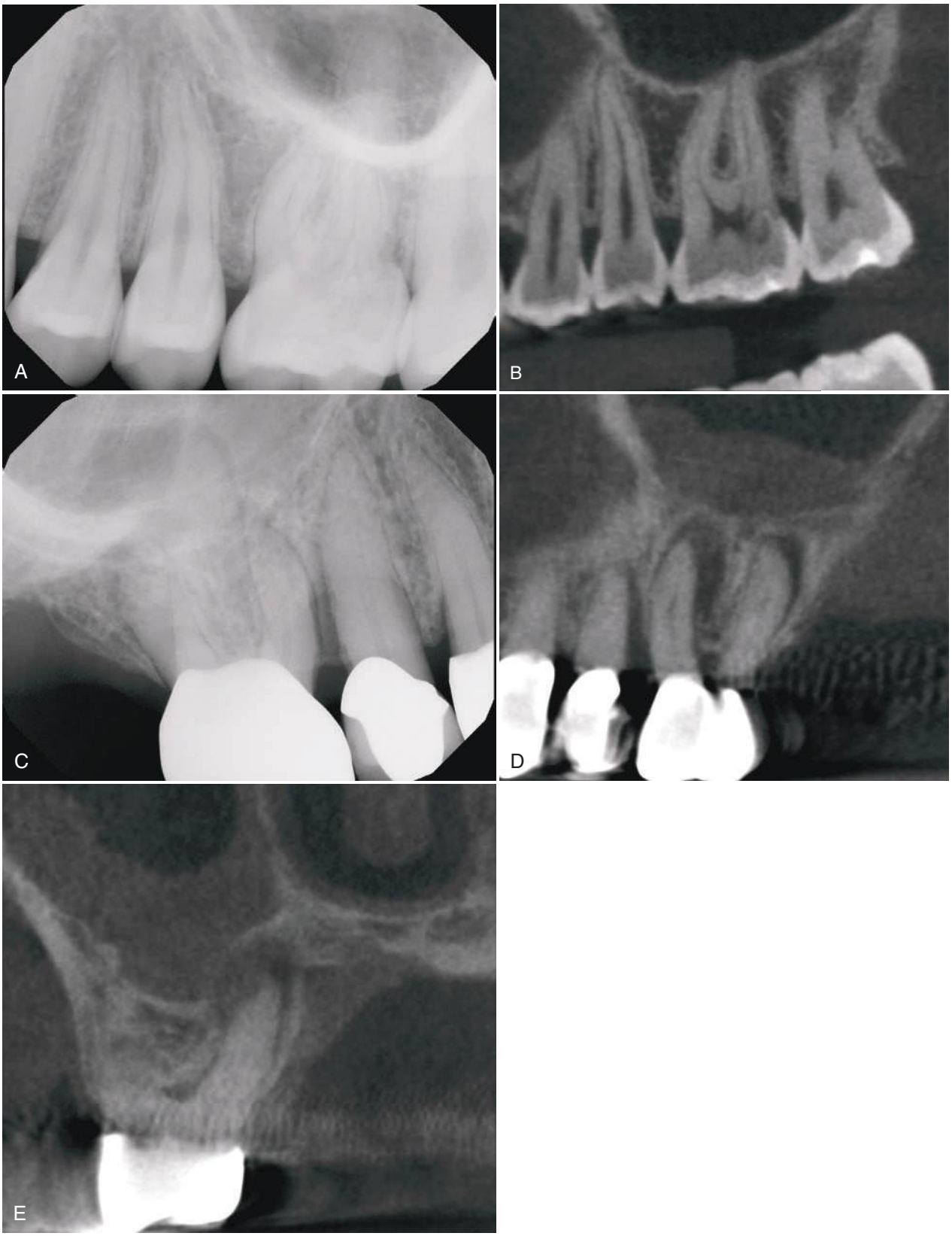


Fig. 3.12 **A,** Periapical image of #14 demonstrating cervical resorption and superimposition of the maxillary sinus over the palatal root apex. There is no way to determine the presence of inflammatory changes in the sinus based on this image. **B,** Sagittal CBCT image obtained because of resorption in first molar. Teeth in the field of view were asymptomatic and responded normally to pulp testing. There is no indication of mucosal thickening in the sinus. **C,** Periapical image of necrotic #3 with well-defined radiolucencies at the apices. **D,** Sagittal CBCT image confirms areas of low density at the root apices and reveals resorption of the maxillary sinus floor with localized mucosal thickening. **E,** Coronal CBCT image illustrates collateral edema into the sinus with a perforation of the sinus floor near the palatal root apex.

PERIAPICAL OSTEOPERIOSTITIS. The inflammatory process from an odontogenic infection can expand the sinus periosteum and displace it superiorly into the sinus. Periapical osteoperiostitis is the osteoblastic reaction that results in new bone formation under the periosteum as it is elevated vertically, resulting in a dome-shaped radiopacity on the sinus floor.[246]

PERIAPICAL MUCOSITIS. Periapical mucositis occurs when odontogenic infections displace the sinus lining without the reactive osteoblastic changes. This will also present as a dome-shaped lesion radiographically, but because the mucosal thickening and tissue edema is limited to the soft tissues, periapical mucositis is more difficult to identify than periapical osteoperiostitis (Fig. 3.13). This type of radiographic change has also been termed *antral pseudocyst*.[90]

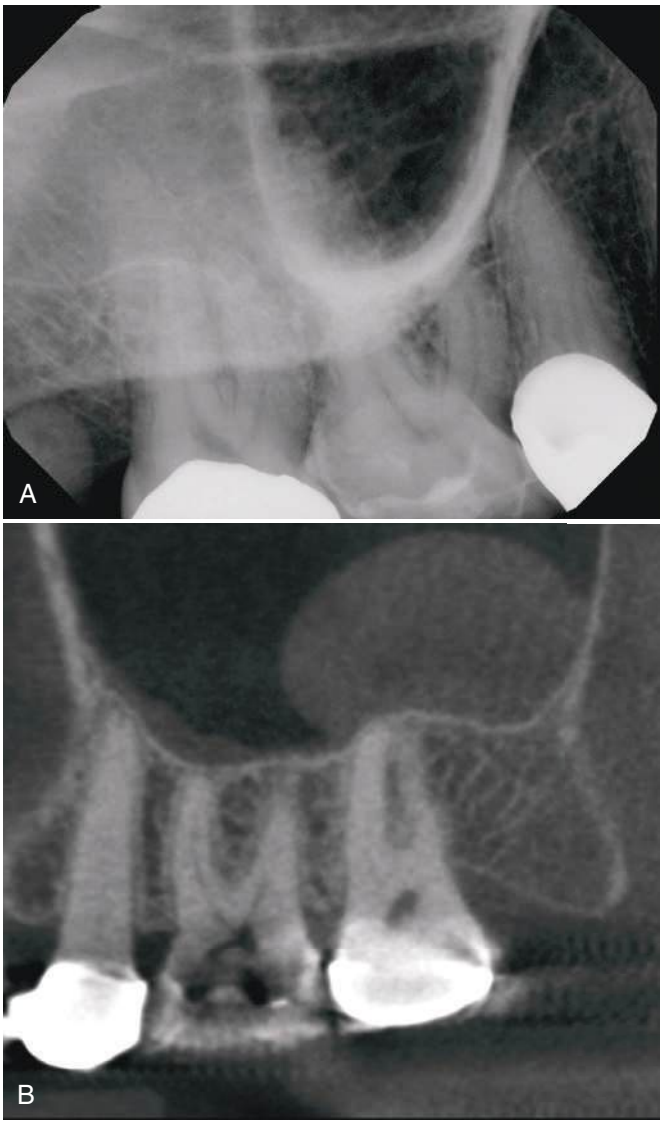


Fig. 3.13 A, Periapical image of #2. This tooth is nonresponsive to thermal testing and painful to percussion. Radiographically there is a slightly increased density in the maxillary sinus near the palatal root apex, but superimposition of normal anatomical structures limits the diagnostic accuracy of the image. **B,** Sagittal CBCT image reveals a homogeneous dome-shaped elevation into the maxillary sinus.

Osteomyelitis With Proliferative Periostitis

Proliferative periostitis is a reactive bone formation in response to the presence of inflammation similar to periapical osteoperiostitis that occurs with MSEO. However, the reactive bone is deposited in parallel rows expanding the surface of the involved jaw, usually in a child or adolescent.[179] It is thought that subperiosteal spread of an inflammatory process elevates the periosteum, stimulating new bone formation.[266] Dental caries with a secondary odontogenic infection is the most common cause. Like condensing osteitis, most cases arise in the premolar and molar area of the mandible, but unlike condensing osteitis, there is obvious expansion, most often of the lower border of the mandible and buccal cortical plates.[128] Removal of the inflammatory stimulus via appropriate endodontic treatment or extraction should result in bone healing over a 6 to 12 month time frame, but biopsy should be considered if bone expansion is present and/or a dental source of infection cannot be identified with clinical examination and pulp testing.

Radiographic Features. Because of superimposition of the lesion over normal bone, proliferative periostitis may be difficult or impossible to visualize with conventional intra-oral radiography (Fig. 3.14). A CBCT or an occlusal radiograph will demonstrate a variable number of layers of reactive bone.[252,274] The boney sheets are typically parallel to each other and the cortical plate. Numerous small radiolucent areas may be identified within the body of the lesion.[66]

Although a radiopacity may be the result of an osteoblastic reaction to an inflammatory stimulus, if pulp testing and clinical history do not indicate an endodontic etiology, other disease entities should be considered in the differential diagnosis. These radiopacities may occur for idiopathic reasons, be developmental in nature, be the result of an underlying bone disorder, or represent variations of normal.

Idiopathic Osteosclerosis

Idiopathic osteosclerosis is typically an incidental radiographic finding characterized by an increased bone density of unknown etiology. The prevalence is estimated to be approximately 5%, but due to inconsistent nomenclature in the literature and the similarity in clinical presentation to condensing osteitis, the true prevalence is unknown.[93,129,198] Idiopathic osteosclerosis is most commonly found in adolescents and may remain unchanged or continue to enlarge slightly until normal growth and development is complete.[198] At that point, the majority of lesions will remain unchanged, but a small number may resolve.[104] The diagnosis of idiopathic osteosclerosis is based on clinical presentation of the lesion, pulp testing of teeth in the area, and restorative history. If any of the clinical findings are suspicious, a biopsy could be done, but typically the diagnosis can be made based on the clinical characteristics and no treatment is indicated.

Radiographic Features. Idiopathic osteosclerosis is identified as an asymptomatic, well-defined, nonexpansile radiopacity (Fig. 3.15). The size of the lesion varies from a few millimeters up to 2 cm, and the outline is often irregular. Like condensing osteitis, it is most commonly located in the first molar region, and there is an overwhelming predilection

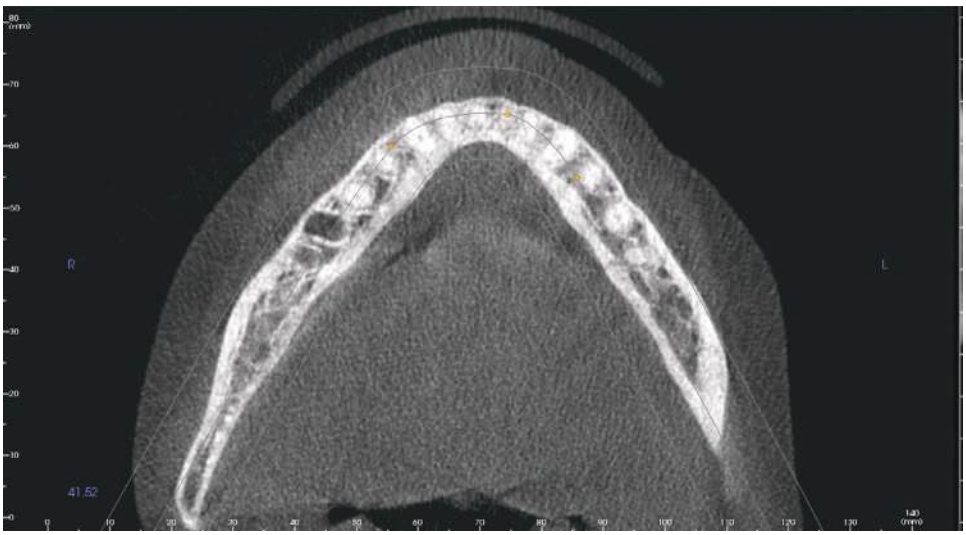


Fig. 3.14 Axial CBCT image of mandibular arch demonstrates irregular medullary bone destruction and cortical defects from tooth #25 to tooth #31. There is new bone formation along the inferior facial cortex consistent with osteomyelitis and proliferative periostitis. (Courtesy Dr. Laurence Gaalaas.)

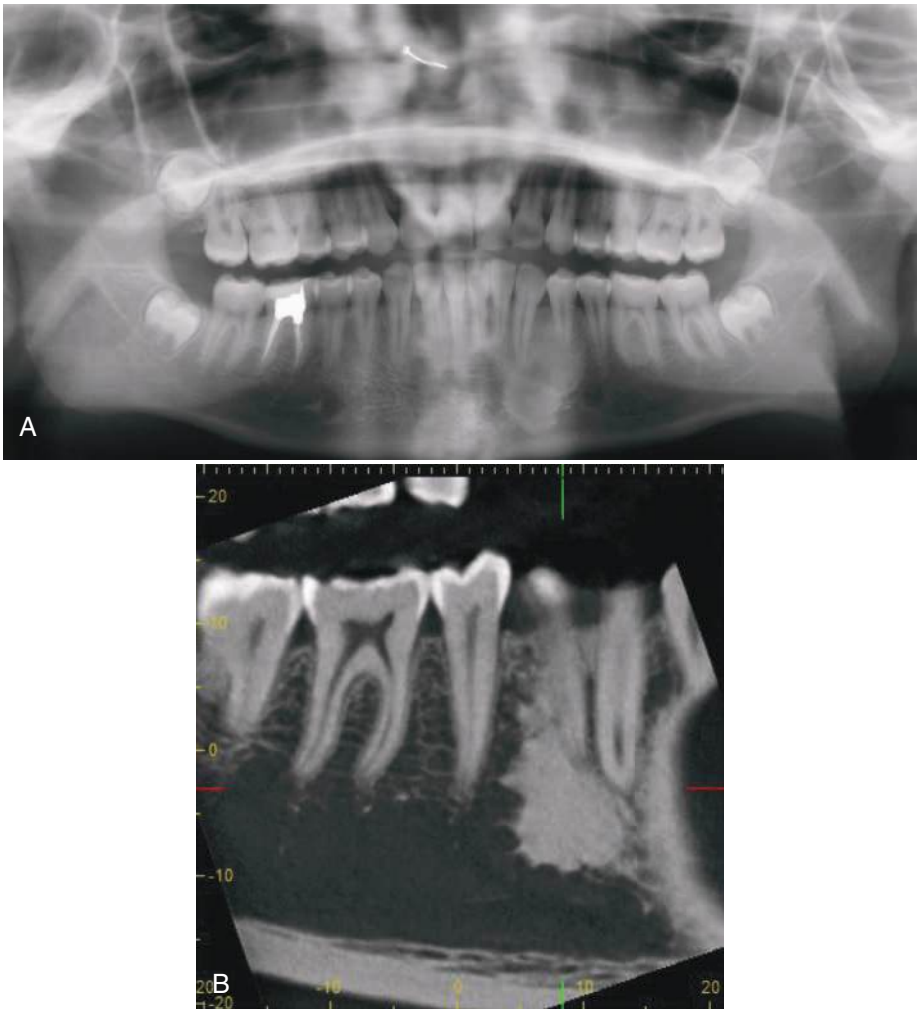


Fig. 3.15 Idiopathic osteosclerosis. **A,** A cropped panoramic image with a well-defined alteration in the trabecular pattern apical to teeth #20 to 22 in a 15-year-old female. Peripherally there is slight cortication with no evidence of root resorption. **B,** Sagittal CBCT image showing a homogeneous area of hyperdensity near the canine and premolar root apices. The lesion has an irregular outline, but there is no indication of root resorption. (Courtesy A, Dr. Laurence Gaalaas; B, Dr. Bruno Azevedo.)

for the mandible.[198,272] It typically presents as a solitary lesion of uniform radiodensity, but multiple radiopacities have been reported in the same patient.[272] Idiopathic osteosclerosis usually occurs near a root apex and root resorption is possible, but uncommon.

Odontoma

Odontomas are common developmental abnormalities and should not be considered neoplasms. They are composed of tissues found in normal teeth, including enamel, dentin, and variable amounts of pulp and cementum. Compound odontomas are made up of multiple structures that resemble teeth and are most often located in the anterior maxilla, while complex odontomas are composed of an unorganized aggregate of enamel and dentin and are most often diagnosed in the molar regions.[37] However, both types will occasionally be found in any maxillary or mandibular site and in the periapical areas.[200] If the diagnosis can't be made based on radiographic findings, surgical removal has an excellent prognosis.

Radiographic Features. Most odontomas are diagnosed before the age of 20 as asymptomatic findings on routine radiographs.[245] Often they are identified when a normal tooth fails to erupt. They are usually smaller than a tooth, but larger odontomas of greater than several centimeters have been reported.[245] Compound odontomas resemble teeth radiographically and are surrounded by a radiolucent border (Fig. 3.16). Complex odontomas have the same radiodensity as teeth and are also associated with a radiolucent border but lack the shape of normal teeth (Fig. 3.17). An early odontoma could resemble a periapical radiolucency if it appears near a root apex prior to calcification.

Paget Disease of Bone

Paget disease is a common metabolic bone disorder characterized by a highly variable and mixed radiographic presentation due to the abnormal and haphazard deposition and resorption of bone. A defect in osteoclasts, and to some degree osteoblasts, results in thickened, enlarged, and weakened bones.[224] It is rare before the age of 40 and tends to affect older white adults with a prevalence that is geographically variable.[179] The highest incidence is found in people of British ancestry, and the prevalence in the United States is estimated to be 1% to 2%.[179] Paget disease has an unknown etiology, and both genetic and environmental factors may be involved. A family history has been reported in as high as 40% of cases, and the decline in overall prevalence over the last 25 years suggests a possible link to a paramyxovirus.[46]

Paget disease is often asymptomatic at the time of diagnosis, but bone pain is not uncommon.[154] Test results indicate an elevated level of serum alkaline phosphatase with normal blood calcium and phosphorus levels. However, alkaline phosphatase is not always abnormal in patients with limited disease. Asymptomatic patients with limited involvement require no treatment and bisphosphonates are used for symptomatic patients to reduce bone turnover and decrease pain. Paget disease tends to be a chronic and slowly progressive disease, but malignant transformation to osteosarcoma is possible.[41]

Radiographic Features. Paget disease typically arises from several bones at the same time. The pelvis, femur, lumbar vertebrae, tibia, and skull are commonly involved.[205,224] The jaws are involved in less than 20% of cases and the maxilla is twice as likely to be affected compared to the mandible. Maxillary involvement is characterized by an

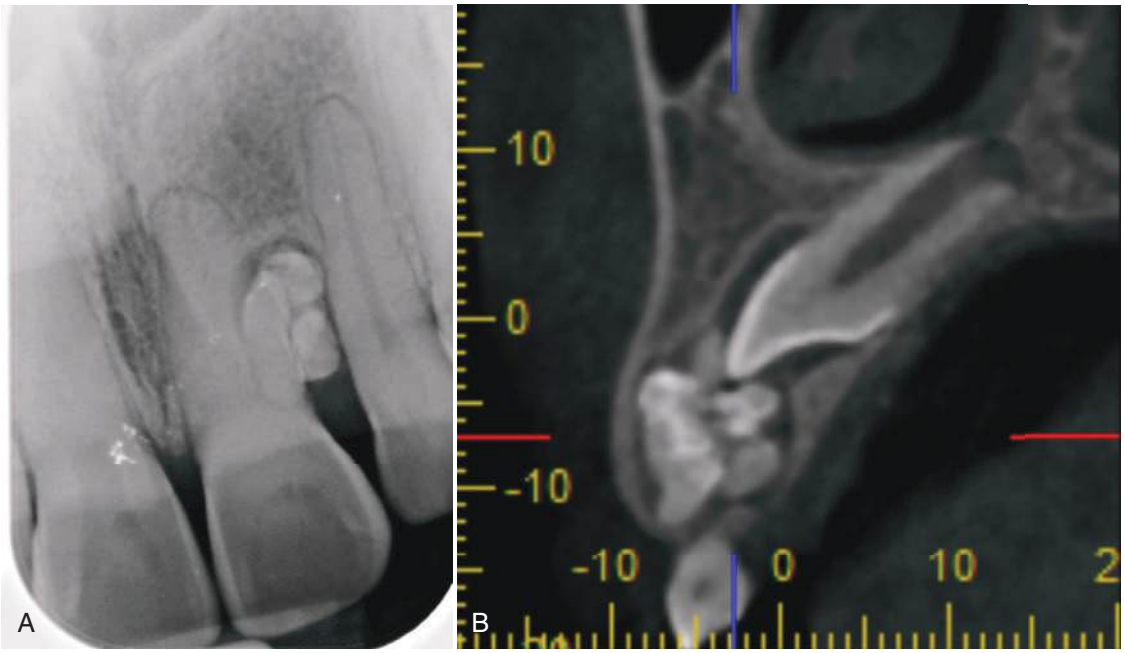


Fig. 3.16 A, Periapical image of a compound odontoma in a 35-year-old female. Multiple radiopacities resembling teeth are located interproximal to the maxillary central and lateral incisors. **B**, CBCT image demonstrating multiple high-density toothlike structures in association with a corticated low-density zone resembling dental follicle. (*A*, Courtesy Dr. Christel Haberland. *B*, Courtesy Dr. Bruno Azevedo.)

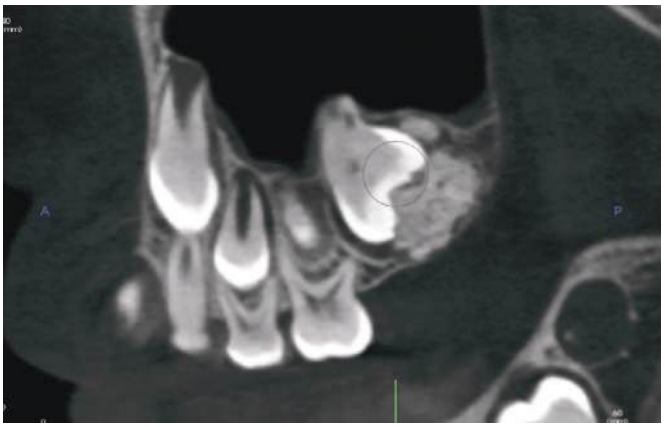


Fig. 3.17 Complex odontoma. Sagittal CBCT image with a high-density globular mass near the crown of a maxillary first molar. The mass exhibits the density of dentin. (Courtesy Dr. Bruno Azevedo.)

enlargement of the midface, and symptoms may include a deviated septum, nasal obstruction, and sinus obliteration. Additionally, enlarged alveolar ridges resulting in space between teeth can occur in either jaw.

Early in the disease process there is decreased radiodensity with a coarse trabecular pattern (Fig. 3.18). During this early resorptive stage, the lesions are very vascular resulting in significant bleeding during oral surgical procedures. Following the resorptive changes, patchy areas of confluent bone sclerosis become more prominent.[39] These sclerotic areas have classically been described as having a "cotton wool" radiographic appearance and may exhibit poor wound healing and an increased risk of osteomyelitis following surgery.[22] Teeth in affected areas may develop hypercementosis with radiographic features similar to cemento-osseous dysplasia.

Exostoses and Tori

An exostosis is a boney outgrowth from the cortical plate. There is a high degree of uncertainty in the literature about the cause and prevalence of exostoses, and they should be considered variations of normal anatomy that have a multifactorial etiology.[86,108,171,189] An exostosis is a relatively common finding on routine clinical examination, but there is minimal literature on the topic and the prevalence within the population is unknown. The most common exostoses are the torus mandibularis and torus palatinus.[179] Exostoses are usually asymptomatic unless trauma results in localized inflammation or ulceration. Treatment is usually unnecessary with the classic clinical presentation. Removal of sizable exostoses may be needed for normal function or for prosthetic reasons.

Radiographic Features. In some cases, the bone density will be high enough that an exostosis will be seen as a radiopacity on dental radiographs. It is rare to see a palatal torus on dental radiographs, but it is not uncommon to visualize a mandibular torus superimposed on the roots (Fig. 3.19).

MULTIFOCAL PRESENTATIONS

Although it is not common to have asymptomatic radiolucencies resulting from simultaneous pulpal necrosis in multiple teeth, it can't be ruled out on radiographic examination alone. Pulp testing and dental history are important if multiple necrotic pulps are suspected (Fig. 3.20), but a multifocal presentation also raises concerns about a nonbacterial etiology.

Periapical Cemento-Osseous Dysplasia

Periapical cemento-osseous dysplasia (PCOD) is the most common multifocal radiolucency to mimic lesions of endodontic origin and has been known by a variety of terms in the literature including, but not limited to, periapical cemental dysplasia, cementoma, and periradicular fibrous dysplasia.[178,227] Like the other fibro-osseous lesions (Box 3.2), PCOD is characterized by the replacement of bone with fibrous and mineralized tissue. PCOD occurs in the anterior mandible and has a predilection for black females.[8,179] The diagnosis is most often made between 30 and 50 years of age and it is uncommon prior



Fig. 3.18 Panoramic radiograph demonstrating Paget disease of bone. A multifocal presentation involving both right and left mandibular molar areas. The coarse trabecular pattern and mixed radiolucency is common to Paget disease. There is significant root resorption, but the typical hypercementosis is absent. (Courtesy Dr. Carl Allen.)

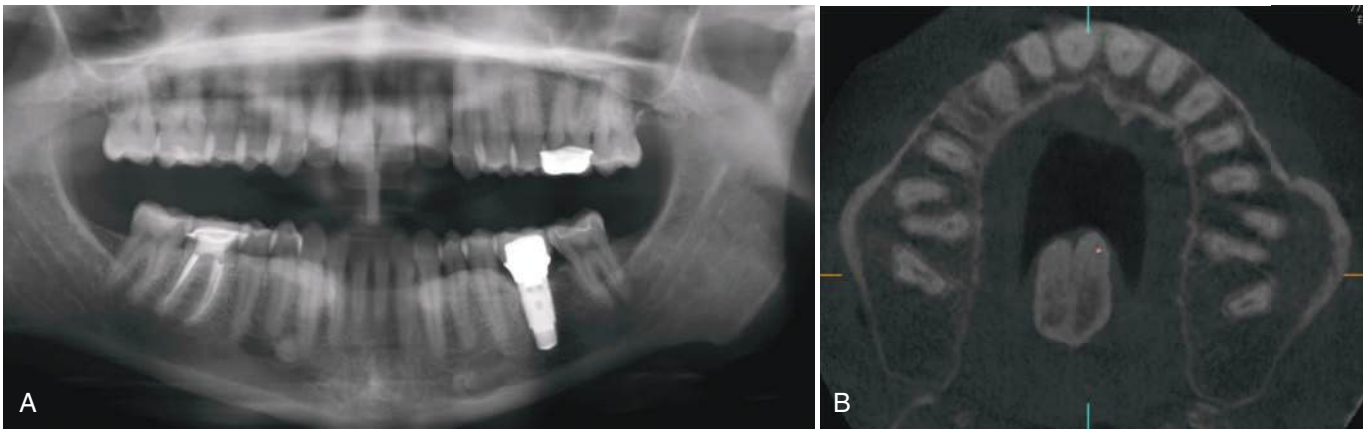


Fig. 3.19 A, A radiodense ridge superimposed over the coronal portion of the roots extending from the implant in the #19 position to the mesial of tooth #30. This represents a prominent lingual exostosis and should not impact endodontic diagnosis but could interfere with visualization of the canals radiographically. This panoramic image also demonstrates well-defined uniform radiopacities inferior to tooth #21 and tooth #27 consistent with idiopathic osteosclerosis. **B,** Axial CBCT image of a maxillary torus characterized by a well-defined high-density area in the midline.

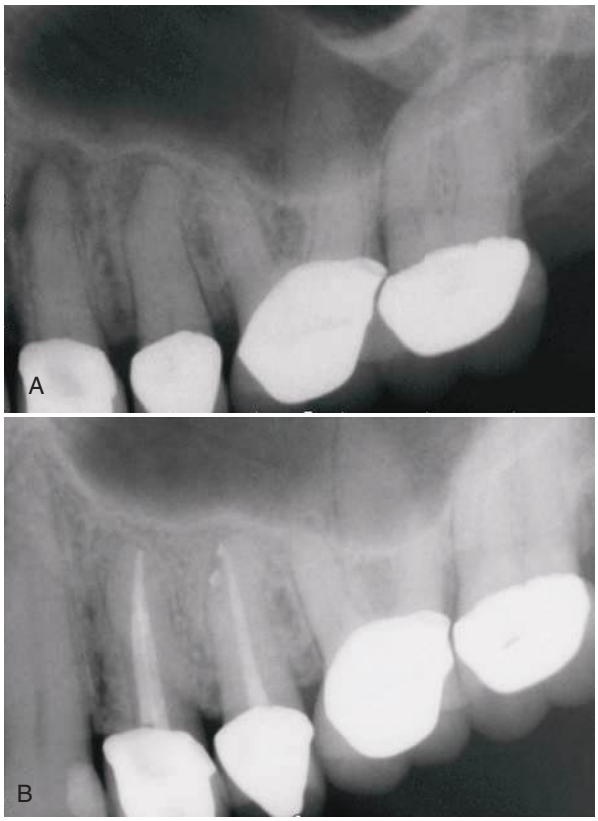


Fig. 3.20 A, Multiple periapical radiolucencies suggest the possibility of a noninflammatory etiology. However, teeth #12 and #13 did not respond to cold testing and both teeth were tender to percussion and palpation. The diagnosis for tooth #12 and tooth #13 was pulpal necrosis and symptomatic apical periodontitis. Nonsurgical root canal treatment was completed in both teeth. **B,** A 6-month recall demonstrated bone healing at the root ends of both teeth.



Fig. 3.21 Multiple well-defined radiolucencies in the anterior mandible. The patient reported no history of trauma and was asymptomatic. These teeth had no significant dental history and responded to both cold and electric pulp testing. Based on this clinical presentation, a diagnosis of periapical cemento-osseous dysplasia was made and no treatment was indicated.

Box 3.2 Fibro-Osseous Lesions

Periapical cemento-osseous dysplasia
Fibrous dysplasia
Focal cemento-osseous dysplasia
Florid cemento-osseous dysplasia
Ossifying fibroma

to the age of 20. The teeth in the area are usually asymptomatic, and the diagnosis is most often made from routine radiographs.[227] The teeth are often unrestored and should respond normally to pulp testing. Treatment is not necessary unless the area is symptomatic.[159] If there is uncertainty about the diagnosis, a biopsy can be obtained, but this comes with an increased risk for infection, bone necrosis, or a disruption in the pulpal blood supply.[15]

Radiographic Features. The radiographic presentation of PCOD is dynamic, variable, and changes as the patient ages.[130,218] Early lesions are mostly radiolucent and will have a strong resemblance to the osteolytic changes that result from pulpal necrosis (Fig. 3.21). As a result, misdiagnosis

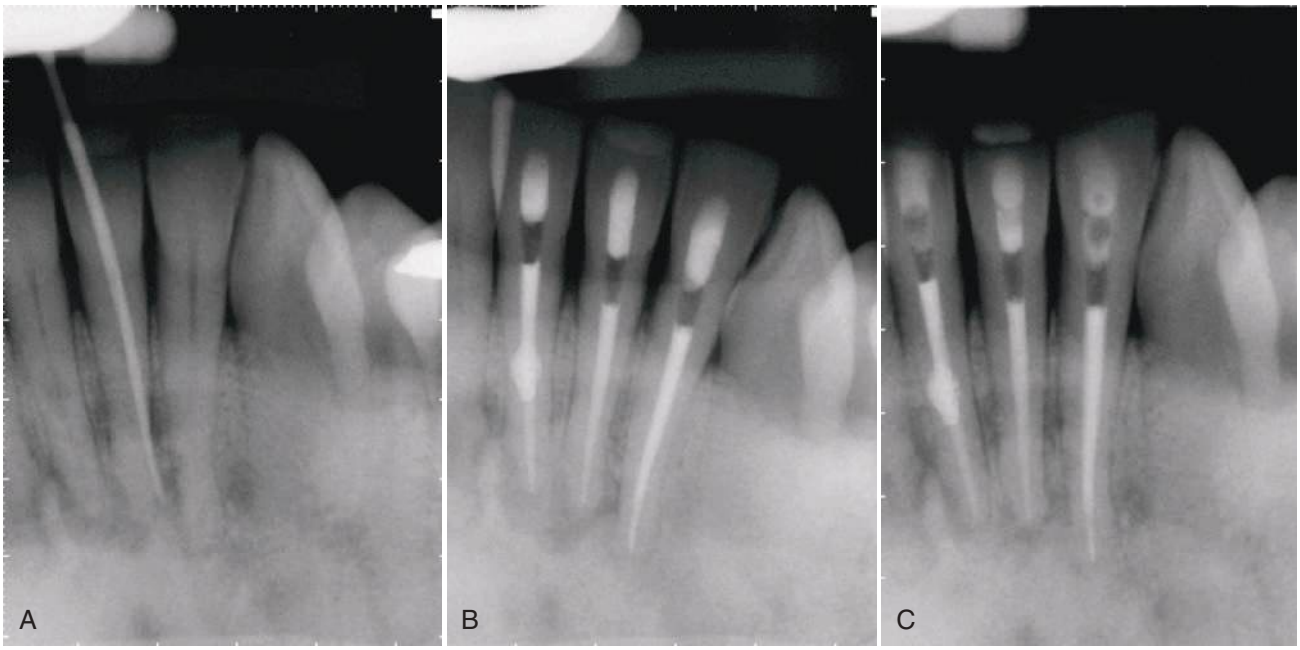


Fig. 3.22 **A,** A 55-year-old female presented with pain, a sinus tract facial to #24, and a history of biopsy in the area several years prior. Teeth #23, #24, and #25 did not respond to cold or electric pulp testing. Radiographically the mixed radiolucent and radiopaque lesions were consistent with periapical cemento-osseous dysplasia, but the sinus tract and pulp testing indicated pulpal necrosis in teeth #23, #24, and #25. Pulpal necrosis was likely due to a disruption of the pulpal blood supply at the time of the biopsy. **B,** Nonsurgical root canal treatment was completed and there was no indication of vital pulp tissue in the canals. The previous biopsy report was obtained prior to root canal treatment, and the diagnosis of periapical cemento-osseous dysplasia was confirmed. **C,** At the 6-month recall the sinus tract had healed and the patient was asymptomatic. Because the radiographic lesions were not the result of a bacterial etiology, there was no expectation to see a significant change in the bone at the 6-month recall.

of PCOD will commonly result in unnecessary root canal treatment. As the lesions change with time, they will take on a mixed radiolucent radiopaque appearance radiographically, and late stage PCOD is characterized by well-circumscribed radiopacities with a well-defined radiolucent border (Fig. 3.22). A multifocal presentation is most common, but solitary lesions that are indistinguishable from periapical cysts or periapical granulomas have been reported.[179] The PDL space is usually unaltered, and even late stage lesions will rarely exceed 1.0 cm.

Florid Cemento-Osseous Dysplasia

Florid cemento-osseous dysplasia (FCOD) most commonly exhibits bilateral, multifocal involvement of the mandible, and all four quadrants can be affected. However, there is no abnormality in blood chemistry.[27,56,233] The lesions can occur in the anterior mandible, but unlike PCOD, the lesions are not limited to the anterior region. Similar to PCOD, there is a predilection for black females, but patients tend to be older at the time of initial diagnosis.[12,56] FCOD can be asymptomatic and discovered on routine radiographs, but pain and exposed bone are also possible.[56,158,222] Because the bone lesions possess a diminished vasculature compared to normal bone, these areas are susceptible to osteomyelitis and the formation of boney sequestra. Tooth extraction and root canal treatment have been reported to initiate secondary infection and symptoms in FCOD.[221] Generally, no treatment is indicated in the asymptomatic patient and the diagnosis can be made based on radiographic findings, clinical examination, and pulp testing.[55] If secondary infection occurs with symptoms, treatment options include antibiotics and surgery.

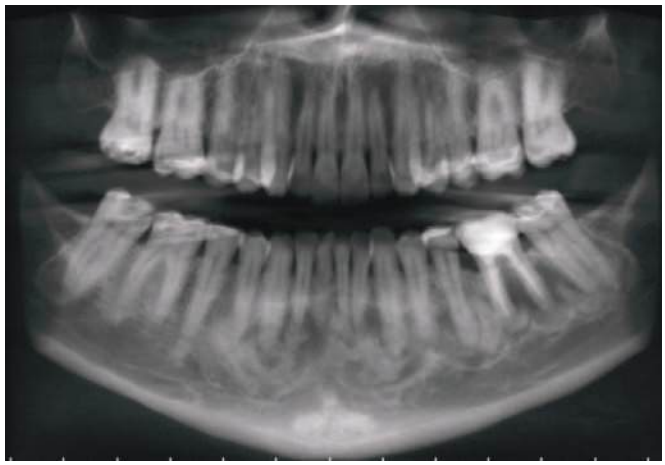


Fig. 3.23 Florid cemento-osseous dysplasia. A mixed radiolucent and radiopaque presentation with multiquadrant involvement. These teeth should respond normally to pulp testing unless there is a history of previous endodontic treatment. If a radiolucent area is present near the root apex of a previous root canal, it may be impossible to determine if the bone loss is due to a bacterial etiology or a manifestation of florid cemento-osseous dysplasia. (Courtesy Dr. Bruno Azevedo.)

Radiographic Features. FCOD exhibits a similar radiographic maturation to that of PCOD. Early lesions are radiolucent and, as they evolve, a mixed radiolucent and radiopaque presentation is common (Fig. 3.23). As patients age, FCOD lesions become more uniformly radiopaque with a thin radiolucent border. These lesions may be well defined, but they are often ill defined, blending with the

normal bone.[56,233] An axial computed tomography (CT) will show high-density masses surrounded by a low-density layer with possible cortical expansion but no perforation of the cortical plate.[12,27] Unlike PCOD, FCOD can fuse with the root apex and blend with the normal bone.[12] A single periapical radiograph may suggest the diagnosis, but a panoramic image would be needed to demonstrate the multiquadrant involvement characteristic of the process.

Hyperparathyroidism

Parathyroid hormone (PTH) is normally produced due to decreased levels of serum calcium. Primary hyperparathyroidism results from an uncontrolled overproduction of PTH and is most often due to a parathyroid adenoma. Primary hyperparathyroidism is usually asymptomatic and can be found with routine testing. These patients have an elevated risk for kidney stones due to elevated levels of serum calcium. Additionally, there is an increased likelihood for the development of duodenal ulcers. Primary hyperparathyroidism is treated by the surgical removal of the parathyroid adenoma or hyperplastic tissue responsible for the excess PTH production.[179,201]

Secondary hyperparathyroidism develops due to chronic overproduction of PTH in the presence of low serum calcium resulting from chronic renal disease. A dietary and pharmacological approach is initially utilized to treat secondary hyperparathyroidism. If this is not successful, the parathyroid glands can be removed or a renal transplant may restore normal vitamin D metabolism.[47]

Radiographic Features. Hyperparathyroidism is characterized by a generalized loss of lamina dura around roots of teeth. Alterations in the trabecular pattern of bone resulting in a "ground glass" appearance tends to be more striking in patients with secondary hyperparathyroidism, and this may be associated with significant bone enlargement clinically.[206] Secondary hyperparathyroidism has also been associated with pain and increased cortical bone density.[3] Additionally, lytic lesions resembling periapical inflammatory disease can occur (Fig. 3.24).[120,155]

The brown tumor of hyperparathyroidism may develop with long-standing disease, especially in female patients with primary hyperparathyroidism.[3] It presents as a well-defined unilocular or multilocular radiolucency. These lesions can be single or multiple and may be located in the mandible, clavicles, ribs, and pelvis.[179,201] A single, well-defined, unilocular brown tumor of hyperparathyroidism on a periapical radiograph would be indistinguishable from a periapical cyst or a periapical granuloma.

Langerhans Cell Histiocytosis

Langerhans cell histiocytosis (LCH) has historically been known by several designations (histiocytosis X, eosinophilic granuloma), and there is controversy about its status as a reactive process or a true neoplasm because LCH also contains a diverse inflammatory infiltrate of eosinophils, lymphocytes, and plasma cells. Additionally, LCH is characterized by a wide spectrum of clinical presentations not typical of a neoplastic process, ranging from single organ involvement that may not require treatment to disseminated and life-threatening systemic disease. However, genetic studies have shown a mutation in an oncogene associated with abnormal cell proliferation in more than 50% of LCH cases, suggesting a true neoplasm.[14] Langerhans cells normally act as antigen presenting cells to T lymphocytes and are located in skin and mucosa. Historically, LCH was thought to arise from these cells due to the presence of Birbeck granules using electron microscopy and immunoreactivity to specific cell surface markers. More recent research has shown that Birbeck granules are not exclusive to Langerhans cells. Currently, there is debate regarding the actual cell of origin that results in LCH with some suggesting that immature myeloid dendritic cells of bone marrow origin are the causative cells.[7,25] Because of its diverse presentation, standardized treatment protocols are limited and ultimately treatment and prognosis are determined by the location and extent of the disease.

Radiographic Features. Patients present over a wide age range, but the majority are diagnosed before the age of 15.[179]

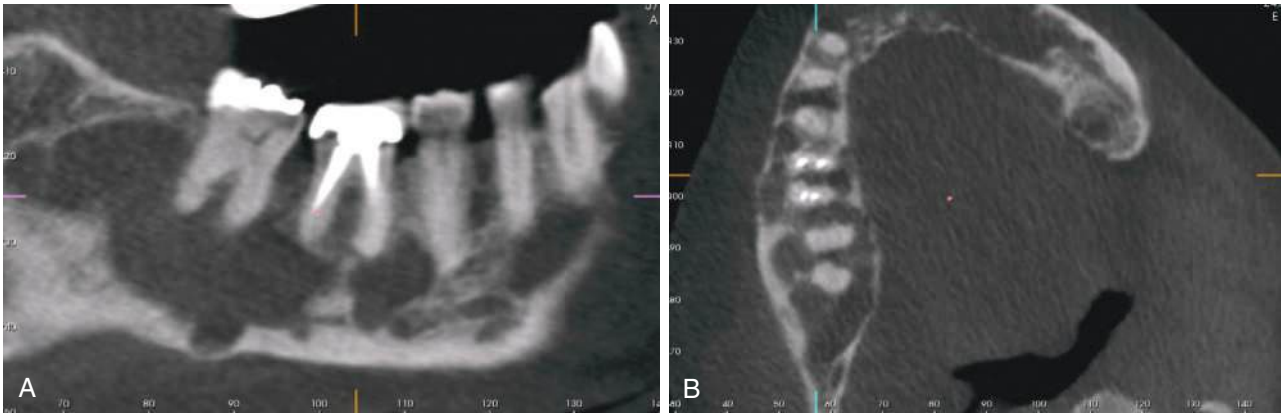


Fig. 3.24 Hyperparathyroidism. **A,** Sagittal CBCT image with multiple low-density lesions and changes in mandibular bone density. **B,** Axial CBCT image showing a thinning of the cortical plates in association with the low-density lesions. (Courtesy Dr. Marcel Noujeim.)

LCH typically presents as single or multiple punched out radiolucencies with well-defined borders, but an ill-defined margin is also possible.[50,112] A solitary well-defined lesion on a periapical radiograph may be indistinguishable from a periapical cyst or periapical granuloma, but the LCH lesion may be more destructive.[196] LCH can have a widespread distribution and affect any bone, but jaw involvement is more common than soft-tissue involvement in the maxillofacial region, and the mandible is affected three times more than the maxilla.[112,179] LCH can result in the loss of alveolar bone and the loosening of teeth, resembling severe periodontal disease (Fig. 3.25).

ILL-DEFINED PATHOLOGIES

At first glance, the peripheral demarcation of a radiolucency may not seem important because periapical cysts and periapical granulomas are the most common ill-defined and well-defined radiolucencies. The differential diagnoses for ill-defined radiolucencies is more ominous and as a result, if pulp testing is not confirmed by the clinical findings at the time of root canal treatment, further investigation and referral would be indicated (Box 3.3). For example, a necrotic pulp would be expected in a tooth with a significant caries history that is nonresponsive to pulp testing and painful to percussion. However, if vital pulp is identified and the patient's symptoms do not resolve following root canal treatment, a biopsy would be recommended if the radiolucent area is ill defined and there is evidence of root resorption, tooth mobility, bone expansion, or perforation of the cortical plates.

Osteomyelitis

Osteomyelitis is an uncommon acute or chronic inflammatory process that spreads through the bone away from the initiating site of infection. In acute cases, symptoms usually present within 2 weeks of the infection while it may take more than 6 weeks to develop symptoms with chronic

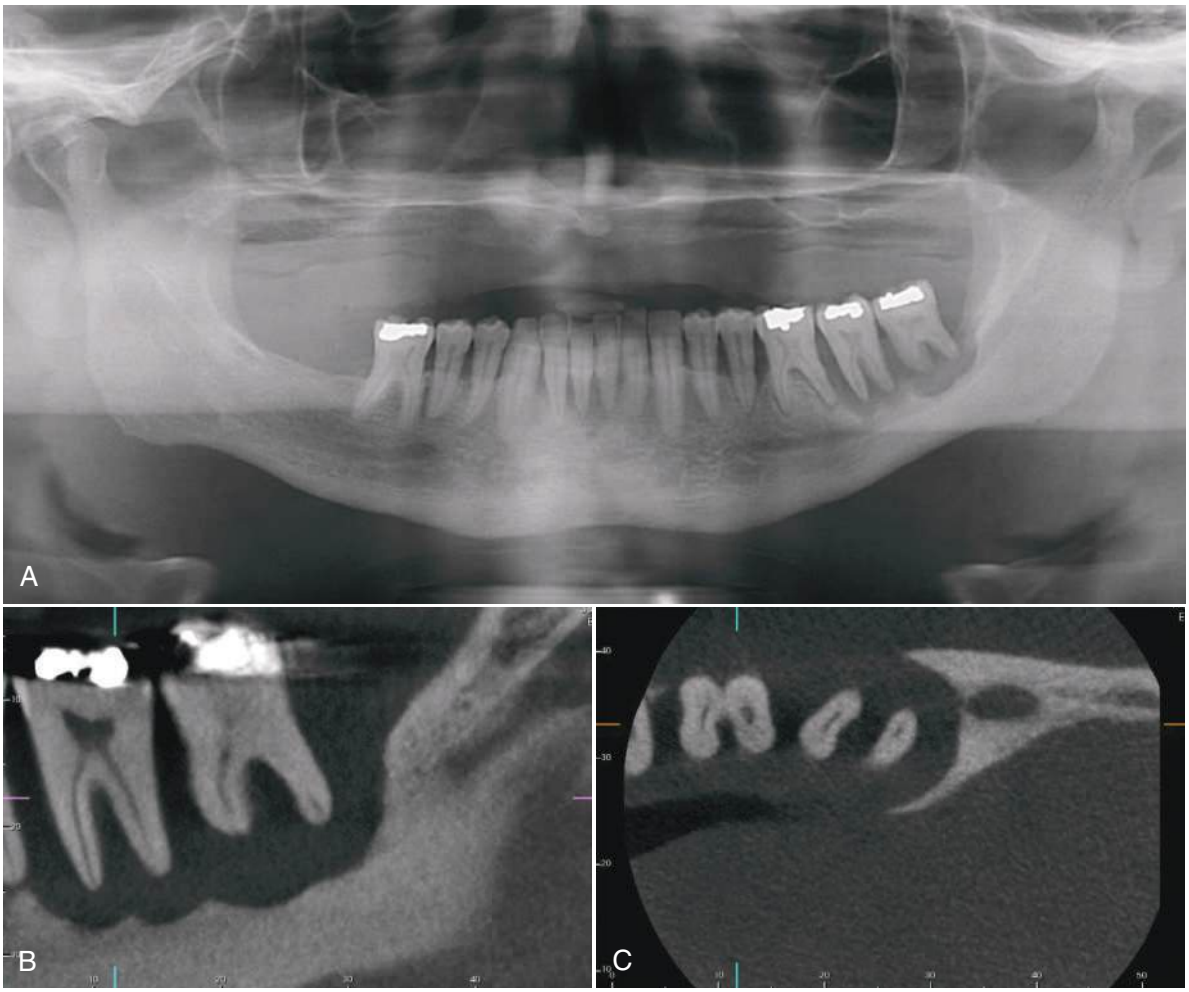


Fig. 3.25 Langerhans cell histiocytosis. **A,** Reformatted panoramic image with destructive radiolucent area in mandibular molar region demonstrating "floating" teeth in the #17 and #18 area. **B,** Sagittal CBCT image confirming the aggressive nature of the lesion. **C,** Axial CBCT image revealing destruction of both buccal and lingual cortical plates. (Courtesy Dr. Marcel Noujeim.)

Box 3.3 Uncommon Ill-Defined Radiolucencies

Osteosarcoma
Chondrosarcoma
Ewing sarcoma
Fibrosarcoma
Odontogenic carcinomas
Salivary gland neoplasms
Osteoradionecrosis
Desmoplastic fibroma

osteomyelitis. Most commonly, a polymicrobial bacterial infection composed of organisms indigenous to the oral cavity serves as the precipitating factor leading to inflammatory bone destruction and the formation of bone sequestra, which are fragments of necrotic bone that are displaced from the viable bone.[140,242] Less commonly, a fungal or mycobacterial infection initiates the osteomyelitis. Predisposing factors include chronic systemic diseases, an immunocompromised state, or any condition that impacts the vascularity of bone. The incidence of chronic osteomyelitis may be increasing due to an increased prevalence of diabetes and peripheral vascular disease in the aging population as well as the increased use of fixation devices in orthopedic surgery.[106] The mandible is more commonly involved due to its more limited vascular supply. Osteomyelitis can occur at any age, and there is a strong male predilection.[140]

Acute Osteomyelitis. Acute osteomyelitis is characterized by the spreading of infection and inflammation through the bone. Patients typically have the signs and symptoms of acute infection, including fever, lymphadenopathy, tenderness to palpation, and elevated white blood cell counts.[242] Paresthesia of the lower lip is possible due to nerve compression, and the extent of swelling is dependent on how the infection spreads.[140] Treatment consists of surgical intervention to remove the source of infection, promote drainage, and remove any infected or necrotic bone. Antibiotics and multiple surgeries may be needed to resolve the infection in some cases, and the emergence of antibiotic-resistant bacteria has served to complicate treatment in an increasing number of cases.[64]

Radiograph changes associated with acute osteomyelitis can be subtle as the infection spreads through the medullary spaces.[124] Periapical images may reveal an ill-defined radiolucency with widened PDL spaces and the loss of lamina dura. As expected, a CBCT will often show the spread of the infection more clearly than conventional dental imaging and, although not practical in the dental setting, magnetic resonance imaging (MRI) can demonstrate edema and inflammatory changes in the marrow spaces characteristic of acute osteomyelitis.[11,13,124] If the infection spreads under the periosteum, there can be an osteoblastic reaction leading to new bone formation with radiopacity or an ill-defined lesion of mixed radiodensity.[105,242]

Chronic Osteomyelitis. Chronic osteomyelitis is the result of an unresolved acute infection causing the inflammatory response to form a layer of granulation tissue around the bacteria. The granulation tissue forms a space that is difficult to access with antibiotics and also serves as a reservoir for bacteria. As seen in acute osteomyelitis, pain is a common presenting symptom.[242] However, swelling and drainage are more likely to be present with chronic osteomyelitis. In advanced cases, tooth mobility and significant bone loss are possible prior to pathologic fracture. Management of chronic osteomyelitis is more difficult due to the pockets of dead bone and infection that are isolated behind the granulation tissue wall.[132] Surgery to remove all infected and necrotic bone is mandatory, and jaw resection may be necessary in some cases. Systemic or intravenous (IV) antibiotics are used in conjunction with surgery. Hyperbaric oxygen has been used in extreme cases, but this is typically not needed if all infected bone has been removed.

The radiographic presentation of chronic osteomyelitis is variable, depending on the duration and extent of the infection and the differing amounts of bone destruction.[273] Chronic osteomyelitis is often characterized by an irregular, ragged, and ill-defined radiolucency. The radiolucent area may contain sequestra and zones of increased density resulting in a mixed radiolucent and radiopaque appearance (Fig. 3.26).[273] Large areas of bone necrosis are possible if blood supply is compromised, and the destructive "moth eaten" nature of some lesions suggests the potential for a malignant neoplasm (Fig. 3.27).[105,116]

Medication-Related Osteonecrosis of the Jaw

Bisphosphonate osteonecrosis of the jaw was first reported when an association was identified between the use of intravenous bisphosphonates and an increased frequency of jaw necrosis.[162,213] However, since these initial reports, other drug classes have been implicated and the name was changed to reflect this in 2014.[214] New medications are frequently being added to the list of causative agents (Table 3.1), and osteonecrosis has been reported in other osseous sites including the small bones of the inner ear and femoral head.[92,166,211]

According to the American Association of Oral and Maxillofacial Surgeons position statement,[214] patients may be considered to have medication-related osteonecrosis of the jaw (MRONJ) if all of the following characteristics are present:

1. Current or previous treatment with antiresorptive or antiangiogenic agents
2. Exposed bone or bone that can be probed through an intraoral or extraoral fistula in the maxillofacial region that has persisted for more than 8 weeks
3. No history of radiation therapy to the jaws or obvious metastatic disease to the jaws

ANTIRESORPTIVE MEDICATIONS. The exact mechanisms leading to MRONJ are unknown and the pathogenesis is likely multifactorial. Osteoclasts are involved in normal bone healing by signaling osteoblasts and influencing angiogenesis. As a result, any medication that impacts osteoclast number or function may influence bone resorption, new bone formation, and the local vasculature. Most cases arise



Fig. 3.26 A, 50-year-old male with a complaint of increasing pain and swelling from the maxillary left canine region over the last year. Examination revealed a palatal swelling and radiographically there was an ill-defined radiolucent lesion near teeth #11 and #12. **B,** A periapical image with a mixed radiolucent and radiopaque lesion is present, but the margins are more well-defined than on the panoramic image. A biopsy was done to rule out a salivary gland neoplasm due to the palatal swelling and destructive nature of the radiolucency. The biopsy revealed both acute and chronic inflammation with sequestrum formation and bacterial colonies consistent with a diagnosis of osteomyelitis. (Courtesy Dr. Laurence Gaalaas.)

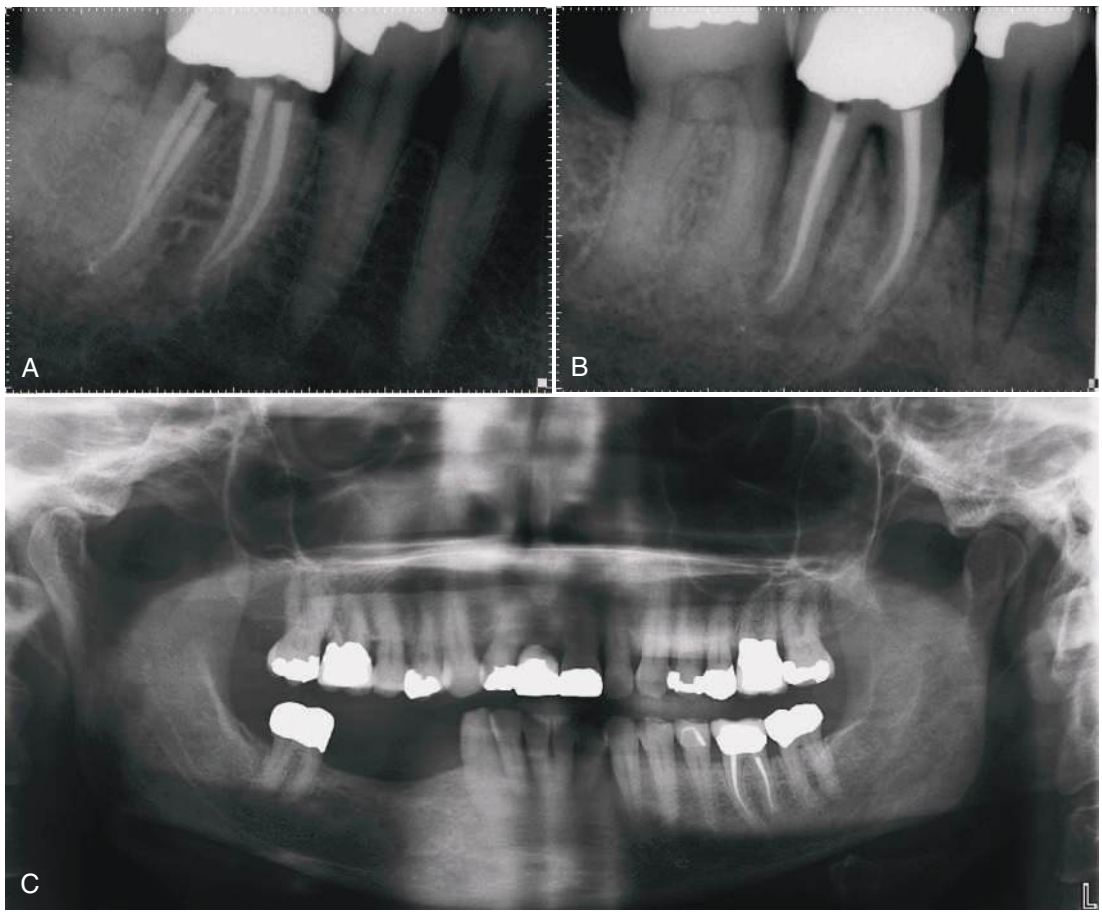


Fig. 3.27 A, Nonsurgical root canal treatment completed in tooth #30 due to pulpal necrosis. **B,** A 1-year recall due to increasing pain in the area. Radiographs revealed an ill-defined radiolucency associated with teeth #28, #29, and #30. These teeth exhibited abnormal mobility and there was significant soft-tissue inflammation, gingival bleeding, and exposed bone. The destructive nature of the bone loss and clinical presentation suggested malignancy, but a biopsy revealed sheets of chronic inflammatory cells and necrotic bone consistent with osteomyelitis. **C,** A panoramic image reveals complete osseous healing at 4-year recall.

Table 3.1 Other Medications Implicated in Pathogenesis of Medication-Related Osteonecrosis of the Jaw

Generic Name	Trade Name
Imatinib	Gleevec
Regorafenib	Stirvarga
Axitinib	Inlyta
Pazopanib	Votrient
Cabozantanib	Cometriq
Dasatinib	Sprycel
Methotrexate	Otrexup, Trexall
Corticosteroids	
Adalimumab	Humira
Infliximab	Remicade
Rituximab	Rituxan
Romosozumab	Evenity
Aflibercept	Zaltrap
Everolimus	Afinitor
Temsirolimus	Torisel
Sirolimus	Rapamune
Radium 223	Xofigo
Raloxifene	Evista

following dentoalveolar surgery, suggesting an inflammatory or infectious etiology could be a contributor to the initiation of MRONJ.[260,271] Periodontal disease and periapical inflammation from a necrotic dental pulp are recognized risk factors for MRONJ, but the risk of developing osteonecrosis in patients exposed to antiresorptive medications following root canal treatment is unknown.[80,214,260,271] Other risk factors for the development of MRONJ include IV bisphosphonate use in cancer patients, corticosteroids, chemotherapy, diabetes, alcohol consumption, poor oral hygiene, and bisphosphonate use for more than 3 years. Smoking and advanced age may increase risk, but the current literature is not definitive.[145,261] In spite of these recognized risk factors, there is no way to reliably predict the development of MRONJ in any individual patient. The use of serum C-telopeptide cross-link of type 1 collagen (sCTX) has been advocated as a biomarker of bone metabolism and a predictor for the development of MRONJ.[163] However, several variables impact sCTX levels, and a systematic review concluded that current evidence does not support the use of sCTX levels as a predictor for the development of MRONJ.[70]

Bisphosphonates suppress osteoclast activity by increasing apoptosis. These medications have an extremely long half-life and exert an effect for many years or decades.[110] MRONJ risk is much higher for cancer patients taking IV doses of bisphosphonates compared to either IV or oral doses for the treatment of osteoporosis. However, the true incidence of MRONJ is not known, and patients with a history of antiresorptive treatment should be made aware of the risks and benefits prior to any treatment linked to the development of jaw necrosis.[9]

Denosumab (Prolia, Xgeva) is a human monoclonal antibody that confers an antiresorptive effect by preventing the development and maturation of osteoclasts. This medication is used both for the treatment of osteoporosis and skeletal-related events due to malignancy. Osteoclasts are derived from the monocyte/macrophage lineage and have a short lifespan of approximately 2 weeks. As a result, osteoclasts need to be continually replenished to keep cell counts constant. Blocking the maturation of these rapidly differentiating cells has a swift and significant impact on the number of functional osteoclasts. Unlike bisphosphonates, denosumab is not deposited in bone and the half-life is less than 1 month, resulting in complete clearance of the drug within 4 to 5 months. In spite of these differences, the risk for the development of MRONJ is very similar for cancer patients taking either denosumab or an IV bisphosphonate.[84,111,241]

ANTIANGIOGENESIS MEDICATIONS. Osteoclasts govern the local vasculature during normal bone healing, and the administration of antiresorptive medications may influence the development of MRONJ by exerting an indirect effect on osteoclast-mediated angiogenesis. However, in some cancer patients, a more direct and profound impact on angiogenesis is desired in an attempt to prevent neovascularization in the tumor or metastatic spread. Sunitinib (Sutent) and sorafenib (Nexavar) are multiple tyrosine kinase inhibitors that block vascular endothelial growth factor (VEGF) and platelet-derived growth factor (PDGF), resulting in decreased angiogenesis, tumor growth, and cancer progression. Bevacizumab (Avastin) is a monoclonal antibody that also targets and inhibits VEGF to limit tumor growth and metastasis. Unlike other initiators of MRONJ, in which the pathogenesis is likely multifactorial, the role of antiangiogenic drugs in the development of osteonecrosis is more intuitive. VEGF and PDGF are involved in normal bone remodeling and wound repair, and the inhibition of their activity leads to a disruption of the normal blood supply, especially in patients undergoing cancer chemotherapy with a compromised immune or healing state. There are multiple case reports linking antiangiogenesis medications to the development of MRONJ.[74,85,115,136,180]

The estimated risk of MRONJ in cancer patients receiving IV bisphosphonates is reported to be 1%, and as stated previously, this is comparable to cancer patients exposed to the RANK-L inhibitor denosumab. The risk for patients receiving VEGF inhibitors is much lower and is estimated to be 0.2% with bevacizumab.[214] However, patients that are treated with combined antiresorptive and antiangiogenic therapies are reported to develop MRONJ at a significantly higher rate, demonstrating the importance of a thorough medical history that includes both current and past medications.[26,43,100]

Treatment options for MRONJ are limited, therefore prevention is key and all patients should have a complete dental screening prior to initiating antiresorptive or antiangiogenesis treatment. For patients undergoing cancer treatment, oral surgery procedures should be completed while allowing several weeks for healing prior to starting medications with

the potential to initiate MRONJ. If a cancer patient has already been exposed to antiresorptive or antiangiogenesis medications, it is best to minimize trauma to the bone, and root canal treatment would be preferred over extraction.[110] If oral surgery is necessary in these patients, the use of a drug holiday has been suggested, but there is little evidence in the literature to support this concept.

Radiographic Findings. Most areas involved with MRONJ are painful due to the presence of necrotic and exposed bone, but this is not universal. The mandible is affected more commonly than the maxilla and both jaws can be involved simultaneously.[217] Panoramic radiographs may show increased radiodensity of the crestal bone in both jaws, and CBCT is better able to demonstrate the sclerotic changes and cortical irregularities commonly associated with MRONJ.[102,256] Prior to necrosis, there is increased radiopacity due to the lack of bone resorption. After the osteonecrosis has been initiated, there will most likely be an irregular, "moth eaten", and ill-defined radiolucent area with or without sequestra.[256] This radiographic appearance can be very similar to that of chronic osteomyelitis (Fig. 3.28). Depending on the location and extent of the necrosis, the radiographic appearance could also resemble periodontal disease or mimic a lesion of endodontic origin. Additionally, with a "moth-eaten" and destructive radiographic presentation, malignancy needs to be ruled out.[89]

Non-Hodgkin Lymphoma

Malignant lymphomas are a heterogeneous group of diseases characterized by a proliferation of lymphoid cells and their precursors. Lymphomas can be divided into two categories: Hodgkin and non-Hodgkin. Hodgkin lymphomas are far less frequent than non-Hodgkin lymphomas and rarely involve extranodal sites. Oral involvement of Hodgkin lymphomas is exceedingly rare and would not mimic lesions of endodontic origin.[267]

Non-Hodgkin lymphomas represent a group of malignancies that most often arise in the lymph nodes and grow as a solid mass, but 40% can arise in an extranodal site outside the lymphoid system.[72,186] In the maxillofacial region, non-Hodgkin lymphomas can result from the spread of disease into the area or they can arise de novo in soft tissue or bone.[186] These lymphomas can originate from either B lymphocytes or T lymphocytes. The incidence is increasing and approximately 70,000 cases are diagnosed each year in the United States with B lymphocytes being the most common cell of origin.[72,134] Patients with immunologic abnormalities have an increased risk for the development of non-Hodgkin lymphoma, and several different viruses have been implicated in the pathogenesis.[157,165]

Non-Hodgkin lymphomas are diagnosed over a wide age range and are more common in adults, but younger individuals can also be affected.[134] The early stage of the disease is characterized by a nontender, slow growing mass that is most likely associated with a group of lymph nodes. This emphasizes the importance of a complete and thorough examination of the maxillofacial region prior to any dental treatment. With time, the number of lesions increases and there may be growth outside of the lymph nodes into surrounding tissues.

Chemotherapy directed against the cell of origin is a common treatment strategy and the prognosis is influenced by the extent of the disease at the time of diagnosis.[165] High-grade lymphomas are treated with chemotherapy and radiation if localized, but radiation will not be used if the disease has spread. The treatment for low-grade lymphomas in older individuals is more debatable. Many of these patients may have no treatment initially, as these lymphomas tend to recur following treatment, and survival of 8 to 10 years without treatment is not unexpected. However, 40% of low-grade non-Hodgkin lymphomas transform to high-grade disease requiring aggressive treatment, and the overall cure rate is low.[179]

Radiographic Features. The radiographic and clinical presentation is variable, depending on the aggressiveness of the tumor at the time of diagnosis. Dull, vague pain that mimics a toothache is present in some cases, while others may have significant pain, swelling, or paresthesia. The maxilla is more commonly involved, but mandibular involvement should not be considered rare.[134] Early lesions



Fig. 3.28 Medication-related osteonecrosis of the jaw. A cropped panoramic radiograph shows a sclerotic rim of bone over the mesial buccal root of tooth #3 and the root apex of tooth #4. There is a well-defined radiolucent area over the midroot of tooth #4 suggesting localized periodontal disease. Purulent discharge and exposed bone were present from teeth #2 to #5. (Courtesy Dr. Christel Haberland.)

may be radiographically subtle and difficult to identify with panoramic imaging.[117] These lesions can resemble a periapical cyst or periapical granuloma and may demonstrate a loss of lamina dura and widening of the PDL space (Fig. 3.29).[61,82,131,138,170,194,236,220] CT and MRI will more clearly reveal bone marrow involvement, often with less cortical destruction than would be expected given the aggressive nature of the neoplastic process. These findings are not specific for non-Hodgkin lymphoma, and other inflammatory and neoplastic diseases need to be considered in the differential diagnosis.[117,164] As the disease progresses, an ill-defined and destructive radiolucency more consistent with a malignant process is likely to emerge. If the lymphoma remains undiagnosed, it will continue to expand and ultimately perforate the cortical plate.

Metastatic Disease

Metastatic deposits of a primary carcinoma can spread via the bloodstream into the jaws, but this is a rare occurrence.[133] Bones that are more commonly involved include the vertebrae, ribs, pelvis, and skull. Although any malignancy has the potential for metastasis, the most common carcinomas to spread to bone are breast, lung, thyroid, kidney, and prostate. Metastatic disease can be asymptomatic or produce pain, swelling, mobility, and paresthesia. The pain level and presentation can mimic pulpal origin.[63,114] In some cases, metastatic disease can precede the diagnosis of the primary tumor, requiring an extensive workup to determine the location of the primary malignancy.[63,113]

Metastatic disease is more likely to occur in older individuals, corresponding to the increased incidence of primary malignancies in adults. However, metastatic disease has been reported over a wide age range. By definition, a metastatic deposit means an advanced stage of disease progression, and treatment is usually palliative. Bisphosphonates have been used to reduce pain levels and limit the risks of a pathologic fracture, but the overall survival rates are poor.

Radiographic Features. Radiographically, metastatic deposits present most commonly as ill-defined radiolucencies in the mandibular molar region that can be destructive leading to pathologic fracture (Fig. 3.30).[63,207] However, metastatic disease can be very subtle in presentation with only a slight widening of the PDL space or, in some instances, the bone may be radiographically normal.[179] It is possible for breast and prostate metastasis to stimulate an osteoblastic reaction locally, resulting in a radiopaque appearance.[96] Depending on location and growth pattern, metastatic deposits can mimic periodontal disease or an endodontic infection radiographically.[59,87,135,225,255]

Multiple Myeloma

Multiple myeloma is an uncommon malignancy of plasma cells that commonly presents with simultaneous involvement in many bones. Multiple myeloma makes up only 1% of all malignancies but is the second most common hematologic malignancy, trailing only lymphoma in incidence.[204] The diagnosis is made using electrophoresis of serum and concentrated urine to demonstrate the presence of M-protein, which is the overproduction of one abnormal immunoglobulin by the malignant plasma cells.[29,203,240]

Multiple myeloma has a mean age at diagnosis of 66 to 70 years and is rarely diagnosed before the age of 40. Most reports suggest there is a slight male predilection, but the incidence for blacks is two times greater than for those of European ancestry.[147] Bone pain is the most common symptom, and this may be associated with fatigue if there is underlying anemia.[71,83] Fever and infection can be present due to neutropenia. A nodular tongue is found in nearly 15% of patients because of amyloid deposition from the abnormal aggregation of M-protein. Additionally, the

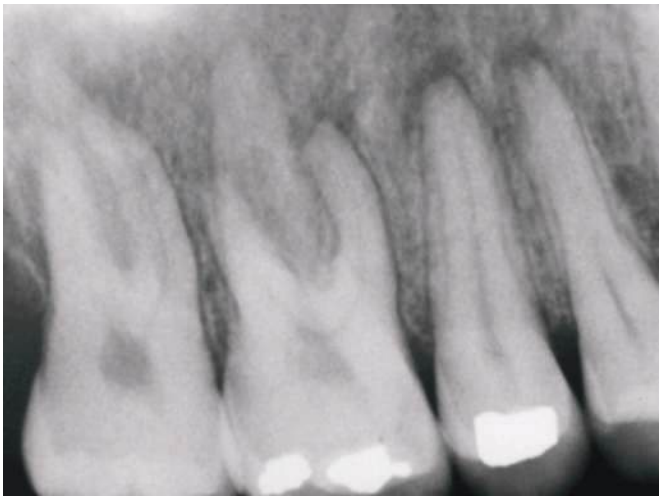


Fig. 3.29 A 29-year-old female with a history of numbness and swelling. These teeth responded normally to pulp testing, but there was a firm and tender buccal swelling from tooth #2 to #6. A diffuse radiolucency was present in the area of the buccal swelling with well-defined radiolucencies at the root apices. Teeth #3 and #4 had limited restorative histories and #2 and #5 were unrestored. Given this clinical presentation a noninflammatory etiology was suspected and biopsy revealed a lymphoma. (Courtesy Dr. Michael Ribera.)



Fig. 3.30 A periapical image demonstrating multiple areas of radiolucency in a 42-year-old female with a chief complaint of achiness and loose teeth in the maxillary right quadrant. These teeth responded to both cold and electric pulp testing. A CBCT revealed an irregular and ill-defined loss of density that perforated the buccal and palatal cortical plates and the maxillary sinus. There was resorption of several root ends. The overall presentation suggested malignancy, and biopsy revealed metastatic breast cancer. (Courtesy Dr. Scott Doyle.)

excessive circulating light chain component of the M-protein can lead to renal failure resulting in proteinuria.[29]

Several chemotherapy regimens have been used historically, but autologous stem cell transplantation has become the first-line treatment for patients deemed healthy enough to tolerate the procedure.[29,204] Radiation for isolated bone lesions is considered palliative, and bisphosphonates are utilized to reduce the likelihood for fracture and pain.[29] Newer chemotherapeutic regimens have resulted in better survival rates, but multiple myeloma is difficult to cure, and treatment often seeks to control disease and prolong survival. Thalidomide, lenalidomide, and bortezomib used in combination with dexamethasone has resulted in improved survival.[144] The prognosis is less favorable for older patients with disseminated disease, and the overall 5-year survival approximates 50% in most reports with a median survival of 6 to 7 years.[204] Medical consultation with the patient's oncologist would be recommended prior to any dental care.[240]

Radiographic Features. Radiographically, there are typically multiple radiolucent areas, some with well-defined, punched-out borders and others that present with more irregular and diffuse margins (Fig. 3.31).[71,149] The presence of multiple lesions suggests a noninflammatory etiology, but a solitary lesion on a single periapical radiograph could mimic a periapical cyst or periapical granuloma. As a result, a panoramic radiograph is recommended if multiple myeloma is suspected.[30] Root resorption is not common but has been reported.[257]

WELL-DEFINED UNILOCULAR PERIAPICAL RADIOLUCENCIES

As stated previously, a unilocular, well-defined apical radiolucency most likely represents a periapical cyst or periapical granuloma if the dental history and clinical testing are consistent with an inflammatory etiology. However, many pathologies, with a wide range of clinical significance, can exhibit the identical radiographic presentation as periapical cysts and periapical granulomas, and the dental literature is replete with case reports of noninflammatory pathologies misdiagnosed as periapical inflammatory disease. The astute clinician needs to consider all possibilities when forming a differential diagnosis.[235]

Periapical Cysts and Periapical Granulomas

Pulp necrosis occurs due to the presence of bacteria within the root canal system.[123] Periapical granulomas represent an accumulation of inflamed granulation tissue as a result of a necrotic pulp. Bacteria and byproducts of bacterial metabolism trigger an immune response, most often at the root apex, but this reaction can occur anywhere along the root surface if the pulp space communicates with the PDL via accessory anatomy or a fracture. The affected tooth should be nonresponsive to pulp testing, unless partial necrosis of a multirooted tooth results in variable levels of responsiveness and a more difficult diagnosis.

The exact mechanism of cyst formation is unknown and numerous theories have been proposed.[175,247,251,253] If epithelium near the root apex of a necrotic tooth is stimulated by an inflammatory response, a periapical cyst can result.[152] There is likely a combination of interactions between the resident tissues in the area, including epithelium, bone, extracellular matrix, and the inflammatory cascade.[24] As

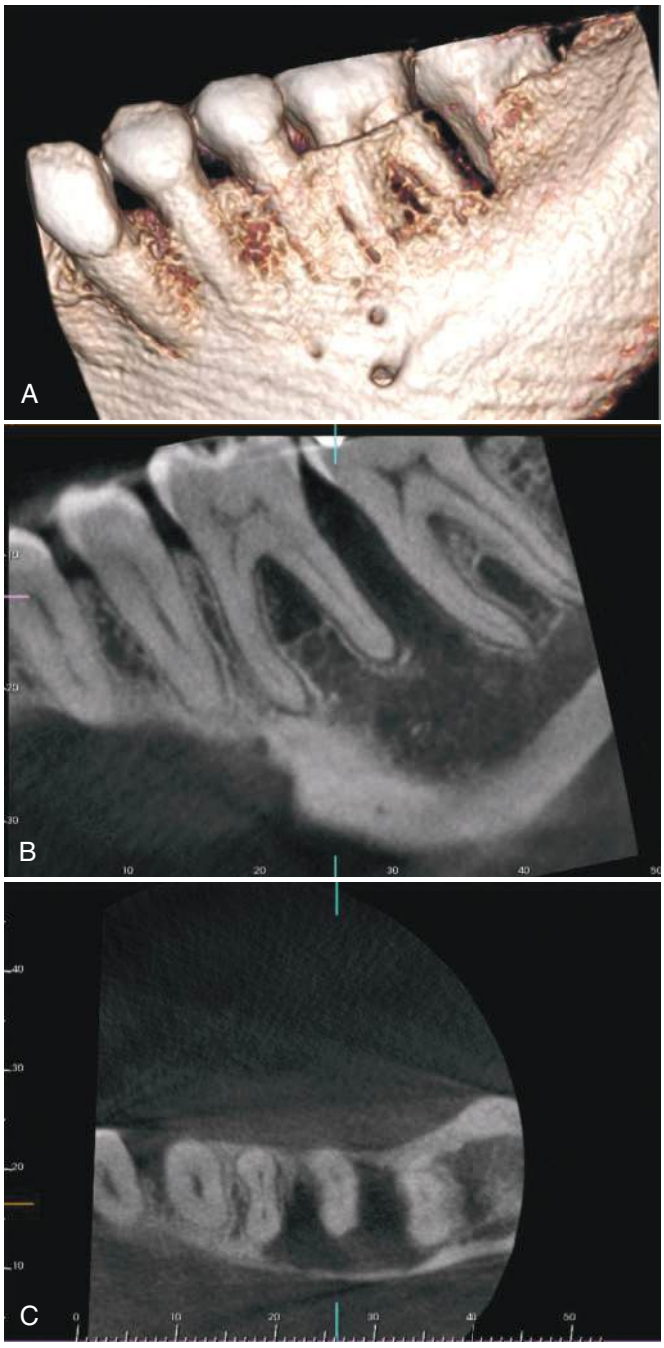


Fig. 3.31 A, Volumetric rendering reveals multiple osseous defects throughout the mandible consistent with multiple myeloma. **B,** Sagittal CBCT image showing a well-defined and destructive low-density lesion. **C,** Axial CBCT image exhibiting destruction of the trabecular bone and marked thinning of both buccal and lingual cortical plates. (Courtesy Dr. Marcel Noujeim.)

with periapical granulomas, the affected tooth does not respond to pulp testing and the diagnosis can be more confusing if partial necrosis is present in a multirooted tooth.

Historically, lesions that were larger than 1.5 cm and well demarcated were considered cysts, and it was thought surgery or extraction would be necessary for complete healing.[88,176] However, this concept has evolved and it is generally accepted that there is no way to differentiate periapical

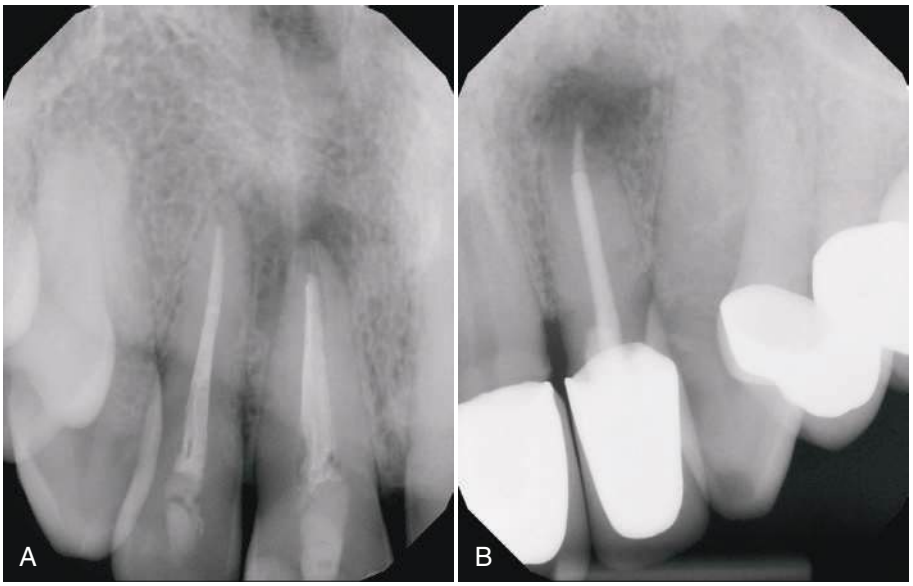


Fig. 3.32 A, Well-defined radiolucency at the root apex of tooth #8. The previous root canal filling is short of the apex and exhibits numerous voids. A tissue sample obtained during apical surgery revealed a periapical granuloma. **B,** Well-defined radiolucency at the root apex of tooth #10. Apical surgery was completed, and the biopsy confirmed a periapical cyst. There is no way to differentiate periapical cysts from periapical granulomas using conventional imaging.

cysts from periapical granulomas using conventional imaging (Fig. 3.32), irrespective of the size or cortication of the radiolucency.[38,208] The ability to differentiate periapical cysts and periapical granulomas utilizing CBCT has also been the focus of numerous studies with inconsistent findings. Guo and colleagues suggested that CBCT is moderately accurate at differentiating periapical cysts from granulomas, while Simon et al. reported CBCT is an accurate differentiator, but found agreement between the pathologist and radiologist in only 76% of cases.[101,232] Other studies concluded that biopsy remains the gold standard to differentiate periapical cysts from periapical granulomas (Fig. 3.33).[35,212]

Multislice-CT has been reported to have a high degree of accuracy distinguishing periapical cysts from periapical granulomas, but this technology requires significantly more radiation exposure and the quality of the image is not as accurate as CBCT.[45,258] Both ultrasound and MRI may be able to differentiate periapical cysts from periapical granulomas, but clinical usage has been limited and these devices are not readily available in the typical dental setting. However, early reports are encouraging and more research is needed to support the use of these technologies.[4,60,153,161]

Because there is no way of knowing whether a radiolucency represents a periapical cyst or a periapical granuloma with imaging alone, it is impossible to know the true incidence of these inflammatory lesions and to subsequently determine if there is a difference in their healing potential following endodontic treatment. Apical surgery or extraction is needed to confirm the diagnosis, and this may lead to a sampling bias of cases that have not healed following routine endodontic treatment. Additionally, in spite of histopathology being considered the gold standard at differentiating periapical cysts from periapical granulomas, this is not without controversy. Some pathologists will make the diagnosis of periapical cyst if an epithelial lining is seen in a histologic section in association with granulation tissue, while other pathologists require visualization of the cyst lining on a basement membrane and an intact luminal surface.

During apical surgery, tissue is curetted from the apex and the sample is typically not removed in a single piece. This may result in a breakdown of the cyst lumen and a decreased likelihood for the diagnosis of a cyst. Moreover, for larger lesions, the entire tissue sample may not be evaluated histologically and slides of representative samples will be viewed by the pathologist. This increases the potential to overlook a small cyst contained in the unviewed tissue. Studies that have utilized intact resected specimens and serial sections to evaluate the entire lesion have reported a lower incidence of cysts compared to many other historical studies.[173,231]

The reported incidence of periapical cysts and periapical granulomas is highly variable in the literature, ranging from 6% to 55%.[17,146,156,174,237] True cysts are completely lined by epithelium and lack any communication with the root canal space. As a result, the healing potential for true cysts may be limited following conventional root canal treatment. In contrast, pocket cysts are in direct contact with the root canal system and may have a greater likelihood for healing following endodontic treatment.[173,231]

Treatment of periapical cysts and periapical granulomas is identical. Both require identification of the offending tooth, allowing removal of the inflammatory stimulus with root canal treatment or extraction. Treatment choices will depend on the patient's treatment plan, periodontal status, restorability of the tooth, and patient desires. The diagnosis of periapical inflammatory disease becomes more complicated if the radiolucency is associated with a previously root canal–treated tooth because pulp testing is no longer relevant.

Radiographic Features. Periapical granulomas and periapical cysts present with variable levels of discomfort, from

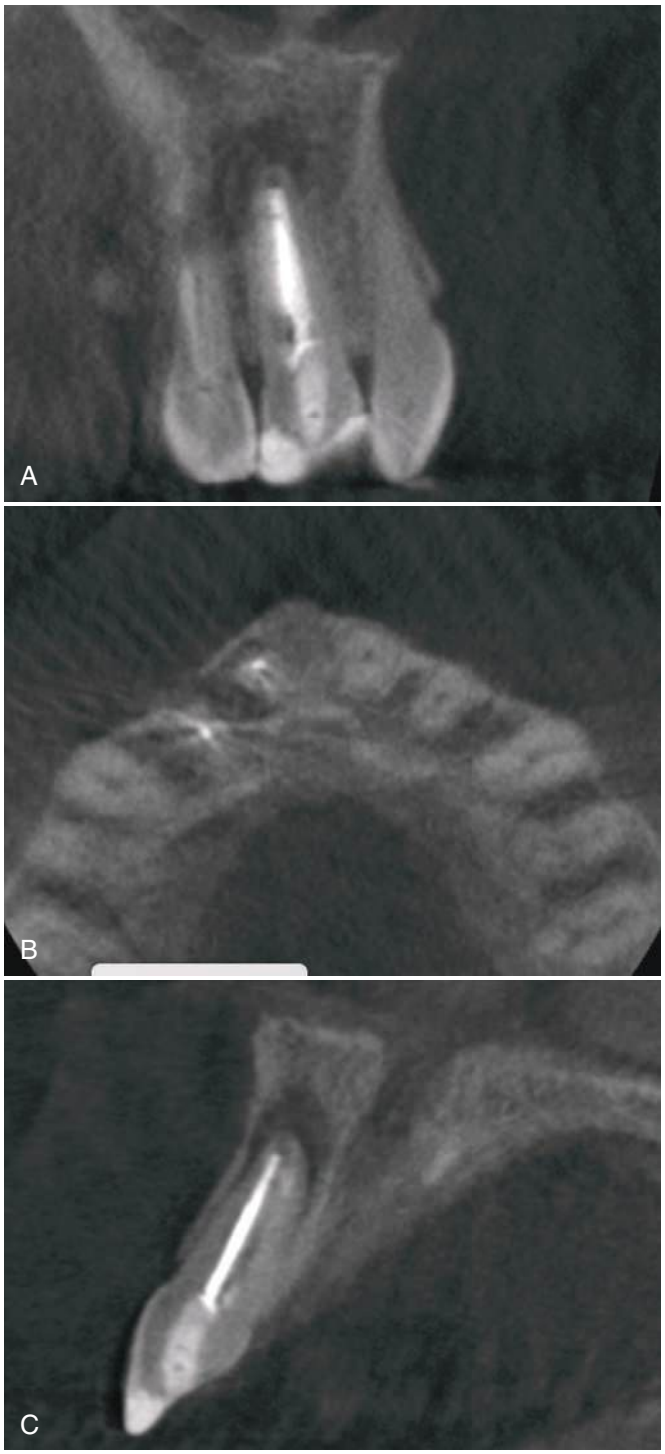


Fig. 3.33 CBCT images of the periapical granuloma from Fig. 3.32, *A*. It is not possible to differentiate periapical cysts from periapical granulomas using CBCT, and biopsy remains the best method to definitively make this distinction. **A,** Sagittal CBCT image of periapical granuloma. **B,** Axial CBCT image of periapical granuloma. **C,** Coronal CBCT image of periapical granuloma.

mild awareness to debilitating pain, and the symptoms often correspond to the pain levels identified with percussion and palpation during clinical examination. Many periapical granulomas and periapical cysts are asymptomatic and are discovered on routine radiographs.

In the earliest stages of infection, neutrophils predominate and there can be very little or no bone involvement, consistent with acute dental pain, but no evidence of abnormality on radiographic evaluation. As the inflammatory cascade persists, there is the potential for more bone destruction and a radiolucent area can develop, ranging from a subtle widening of the PDL space to extensive bone loss of more than several centimeters (Fig. 3.34). The radiolucent area can exhibit a well-defined and corticated margin or the inflammatory process can stay within the medullary spaces and lack definition radiographically (Fig. 3.35). Root resorption is not uncommon and the outline and definition of the radiolucency can be altered depending on the angle used to obtain a two-dimensional periapical radiograph. As a result, it is often necessary to obtain multiple radiographs to identify a subtle periapical lesion, but this may not improve diagnostic accuracy in all cases.[36] A CBCT may be a better alternative if there is doubt about the presence of periapical pathology.[142]

Lateral Radicular Cyst

The bacterial etiology of a lateral radicular cyst is identical to that of a periapical cyst or periapical granuloma; the only difference is in the location affected. As a result, pulp testing is critical to determine the causative tooth because several other pathologies can look radiographically identical in this area. Endodontic treatment or extraction is indicated depending on the patient's treatment plan, periodontal status, restorability of the tooth, and patient preference.

Radiographic Features. Lateral radicular cysts are most commonly well-defined radiolucencies, and some may have a corticated border (Fig. 3.36). However, these lesions will exhibit the same growth patterns as periapical cysts and periapical granulomas resulting in an ill-defined radiolucency when the inflammatory cascade is confined to the medullary spaces. A CBCT will better demonstrate the radiographic changes when the lesion is small or lacks definition. The differential diagnosis for radiolucencies in the lateral location includes, but is not limited to, the following: OKC, ameloblastoma, and lateral periodontal cyst (LPC).

Lateral Periodontal Cyst

The LPC is an uncommon developmental cyst of odontogenic origin thought to arise from rests of dental lamina. Although the LPC is developmental and occurs over a wide age range, diagnosis is uncommon before 30 years of age. The mandibular canine and premolar areas are the predominant sites of involvement, but reports indicate that it can occur in either arch.[40,79,234] LPCs are often discovered as an asymptomatic finding on routine radiographs, but expansion, tenderness to palpation, and pain are possible. Most present as a single lesion, but LPCs can have a multifocal presentation, and a multicystic variant known as a botryoid LPC has also been described.[127,263] Surgical removal is the treatment of choice and recurrence is not likely. However, due to the multicystic nature of the botryoid subtype, a higher recurrence rate is possible.[103,199]

Radiographic Features. Radiographically, the LPC is characterized by a unilocular, well-defined, oval-shaped

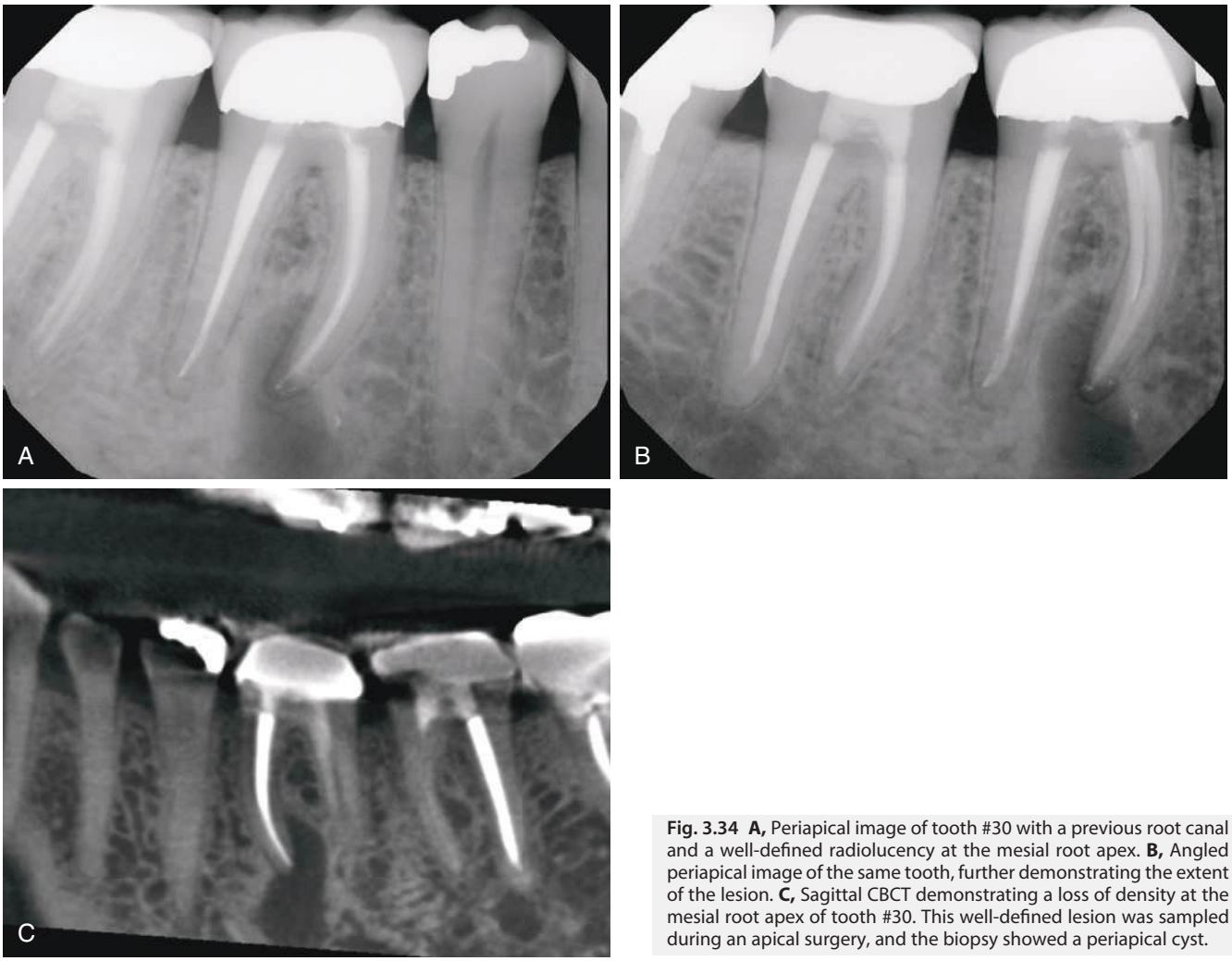


Fig. 3.34 A, Periapical image of tooth #30 with a previous root canal and a well-defined radiolucency at the mesial root apex. **B,** Angled periapical image of the same tooth, further demonstrating the extent of the lesion. **C,** Sagittal CBCT demonstrating a loss of density at the mesial root apex of tooth #30. This well-defined lesion was sampled during an apical surgery, and the biopsy showed a periapical cyst.

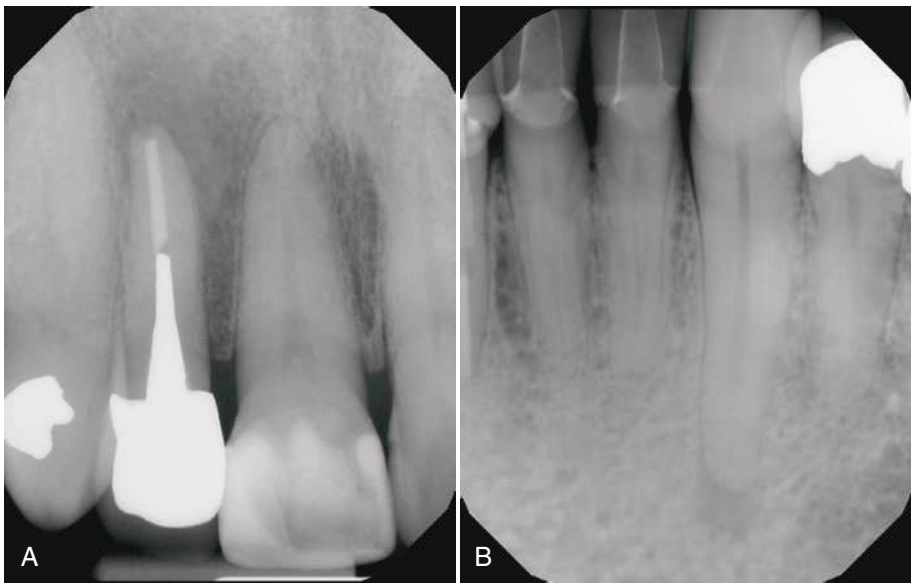


Fig. 3.35 A, Biopsy revealed a periapical cyst at the apex of tooth #7. There is a loss of trabeculation at the apex blending into normal bone resulting in an ill-defined radiolucency. **B,** Well-defined periapical granuloma at the root apex of tooth #27 was confirmed with a biopsy when bone loss and symptoms persisted following nonsurgical root canal treatment. Size and radiographic definition of periapical cysts and periapical granulomas are highly variable.

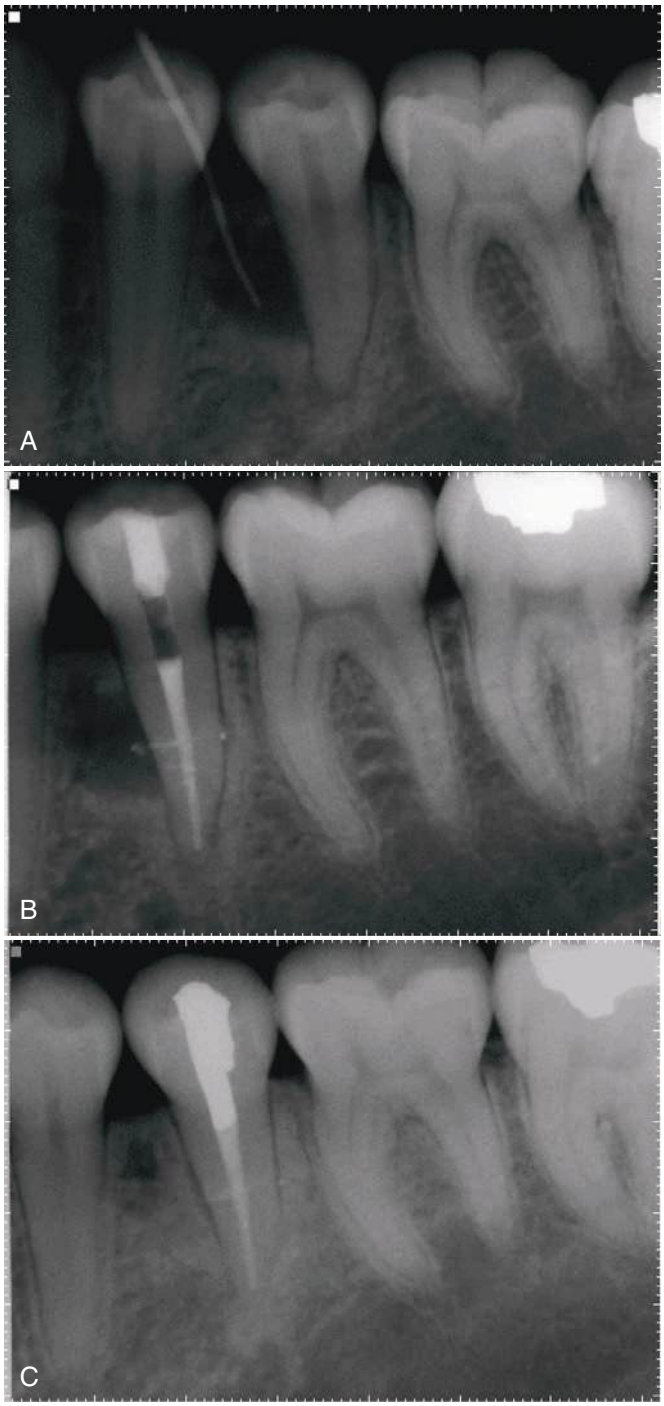


Fig. 3.36 Lateral radicular cyst. **A,** Preoperative image with gutta-percha tracing a draining sinus tract. Patient has a history of biopsy in the area that was diagnosed as a dentigerous cyst. Tooth #20 responds to both cold and electric pulp testing. Options included root canal treatment, extraction, or a second biopsy. The sinus tract suggests a necrotic pulp and the potential for a false-negative pulp test. Also, the diagnosis of dentigerous cyst is not accurate because the tooth has erupted. Nonsurgical root canal treatment was completed. **B,** Several lateral canals can be seen opposite the radiolucent areas and the sinus tract resolved within 1 week. **C,** Healing of bone is evident at 18-month recall.

radiolucency that is indistinguishable from lateral radicular cysts in many cases (Fig. 3.37). However, the adjacent teeth should respond to pulp testing, unless there is pulpal pathology for unrelated restorative reasons. This radiographic presentation is also very similar to early unilocular OKCs or ameloblastoma (Fig. 3.38). Because they are slow growing, most are less than 1 cm at the time of diagnosis. Less commonly, an LPC can be located near a root apex, suggesting a lesion of endodontic origin.[181] The botryoid subtype can present as a multilocular radiolucency, but it is most often unilocular in spite of its multicystic growth pattern.

Focal Cemento-Osseous Dysplasia

Focal cemento-osseous dysplasia is another fibro-osseous lesion characterized by replacement of bone with fibrous and mineralized tissue. As the name implies, focal cemento-osseous dysplasia involves a single site, unlike PCOD, which is more commonly multifocal. Although the exact etiology is unknown, the proximity to root ends and a cementum-like component suggest the cells of origin reside in the PDL.[244] As with PCOD, the majority of cases affect females and are diagnosed after 30 years of age.[130,244] Although there is a predilection for the black population, it is not as strong as is seen with PCOD. The diagnosis could be based on radiographic findings and pulp testing, but other pathologies, with more significant growth potentials, can resemble FCOD, necessitating a biopsy for definitive diagnosis. After the diagnosis has been established, routine follow-up of the affected bone is indicated.

Radiographic Features. Focal cemento-osseous dysplasia usually involves the posterior mandible, is commonly asymptomatic, and is discovered on routine radiographic examination.[159,244] Most lesions are smaller than 1.5 cm and exhibit a variable and mixed radiolucent radiopaque appearance, but some may be completely radiolucent (Fig. 3.39). There is usually a well-defined border, but not always. They tend to occur near root ends and will mimic periapical inflammatory disease if no radiopacity is present within the lesion.[62]

Nasopalatine Duct Cyst

Nasopalatine duct cysts, also referred to as incisive canal cysts, are the most common nonodontogenic cyst in the jaws with a prevalence of approximately 1%.[177,179] They are likely developmental but are most often diagnosed after the age of 30, suggesting a possible traumatic or bacterial etiology. The diagnosis is typically based on radiographic findings, and nasopalatine duct cysts have an excellent prognosis following surgical removal.

Radiographic Features. Nasopalatine duct cysts can occur at any age, but they are more likely found in adults and are rare before 10 years of age.[107,223] Swelling, drainage, and pain are the most common symptoms associated with these cysts. Many are asymptomatic and discovered on routine radiographs. They present as a well-defined, round or heart-shaped radiolucency in the maxillary midline or just lateral to the midline (Fig. 3.40). Most are between 1 cm and 2 cm, but they will uncommonly exceed 5 cm and

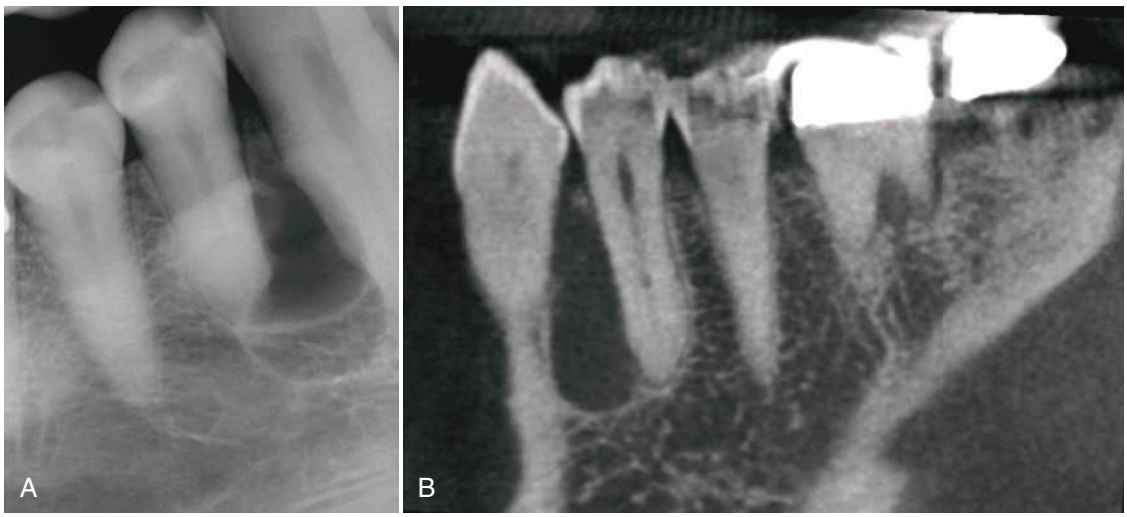


Fig. 3.37 Lateral periodontal cyst. **A,** Periapical image of a well-defined radiolucency near the root apex of mandibular premolar and canine. The periapical image suggests a multilocular presentation that would not be consistent with periapical inflammatory disease from a bacterial etiology. **B,** Sagittal CBCT image revealing the loss of density between the canine and premolar teeth with a normal PDL space at the apex of tooth #21. (Courtesy Dr. Martin Rogers.)

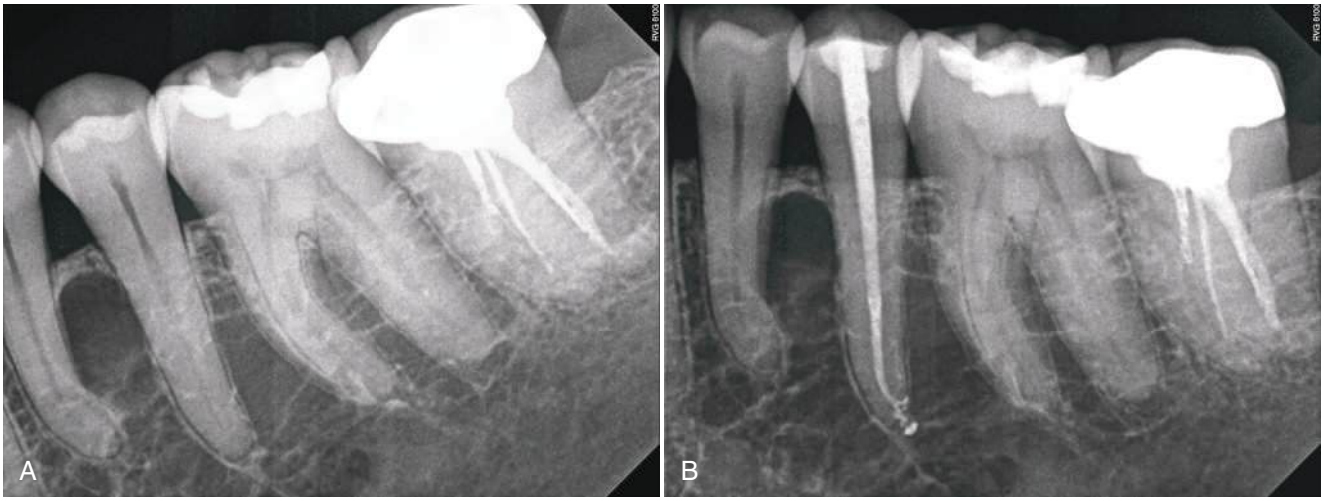


Fig. 3.38 **A,** Pulp testing indicated irreversible pulpitis symptoms from tooth #20 and a well-defined radiolucent area was identified between teeth #20 and #21. **B,** Nonsurgical root canal treatment was completed to address the thermal symptoms, and referral was made for a biopsy that identified an odontogenic keratocyst. Lateral radicular cysts, unicystic ameloblastomas, odontogenic keratocysts, and lateral periodontal cysts can exhibit identical radiographic appearances in the mandibular canine and premolar areas. (Courtesy Dr. Karen Potter.)

exhibit a more destructive nature.[195,215] Due to superimposition over the root ends of central incisors, pulp testing and clinical correlation are necessary to avoid mistaking nasopalatine duct cysts for lesions of pulpal origin.[78]

Stafne Defect

Stafne defects are asymptomatic radiolucencies most commonly located near the angle of the mandible, but a less common anterior position has been reported.[34,52,126,239] The exact etiology is unknown and it is often considered to be developmental in nature. However, the radiolucent area is usually not diagnosed until middle age, leading some investigators to postulate pressure resorption as the cause. There is a strong male predilection and due to the classic radiographic presentation, the diagnosis of posterior defects is typically made based on radiographic findings alone. No treatment is indicated unless there is uncertainty about the diagnosis, and biopsy most often yields normal salivary gland tissue.

Radiographic Features. Because Stafne defects are asymptomatic they are most often discovered as an incidental finding on panoramic radiographs.[148] They present as radiolucencies on the lingual side of the mandible inferior to the mandibular canal with a well-defined and corticated border (Fig. 3.41).[262] Although the size is variable, most Stafne defects are between 1 cm and 2 cm and, in some instances, the radiolucent area will

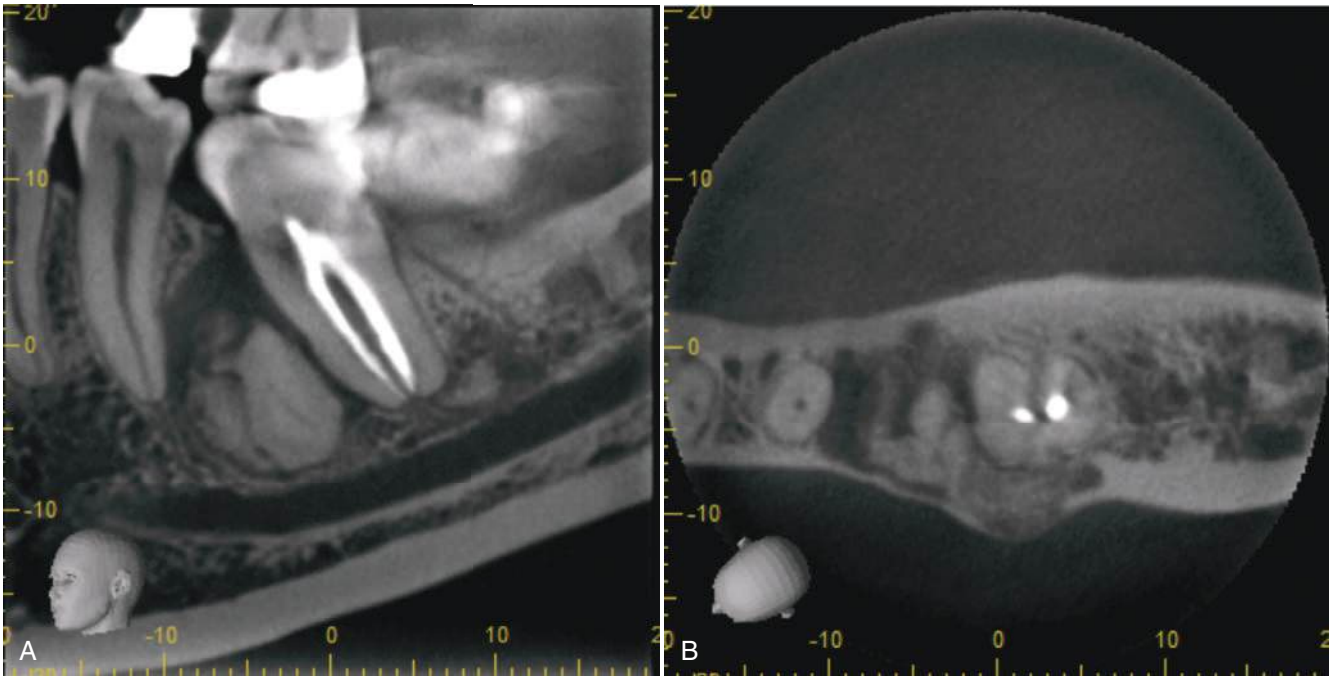


Fig. 3.39 Focal cemento-osseous dysplasia. **A,** Sagittal CBCT image demonstrating a mixed-density lesion with a low-density periphery between the mandibular second premolar and first molar. A second high-density lesion is present near the distal root apex of the first molar. **B,** Axial CBCT image exhibiting an expanded and thinned cortex from a mixed-density central globular mass with a hypodense periphery. (Courtesy Dr. Bruno Azevedo.)



Fig. 3.40 **A,** Periapical image with a well-defined unilocular radiolucency at the root apex of tooth #8. Asymptomatic and no history of trauma. Tooth #8 responded to both cold and electric pulp testing. A biopsy revealed a nasopalatine duct cyst. **B,** The more classic presentation of nasopalatine duct cyst with a rounded radiolucency in the midline. (Courtesy A, Dr. Christel Haberland; B, Dr. Martin Rogers.)

approximate the root ends of the mandibular teeth, suggesting a periapical cyst or periapical granuloma (Fig. 3.42). Anterior Stafne defects associated with the sublingual gland are less common. Superimposition over anterior root ends and the atypical position suggest periapical inflammatory disease or any other unilocular well-defined radiolucency. In these cases, it is important to correlate clinical and restorative histories with the radiographic findings to ensure appropriate treatment.[58] Additionally, both the axial and coronal planes of a CBCT will demonstrate the lingual concavity and lack of involvement at the root ends.

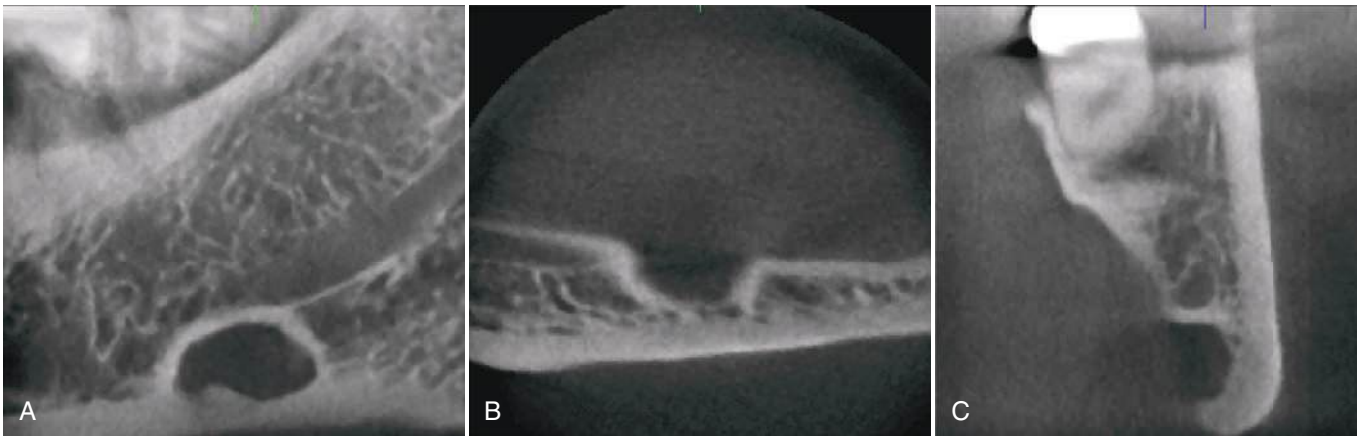


Fig. 3.41 Stafne defect. **A,** Sagittal CBCT image showing a well-defined loss of density inferior to the mandibular canal. **B,** Axial CBCT image demonstrating a lingual concavity. **C,** Coronal CBCT image also exhibiting a lingual concavity inferior to the mandibular canal with no involvement of the root ends. (Courtesy Dr. Bruno Azevedo.)

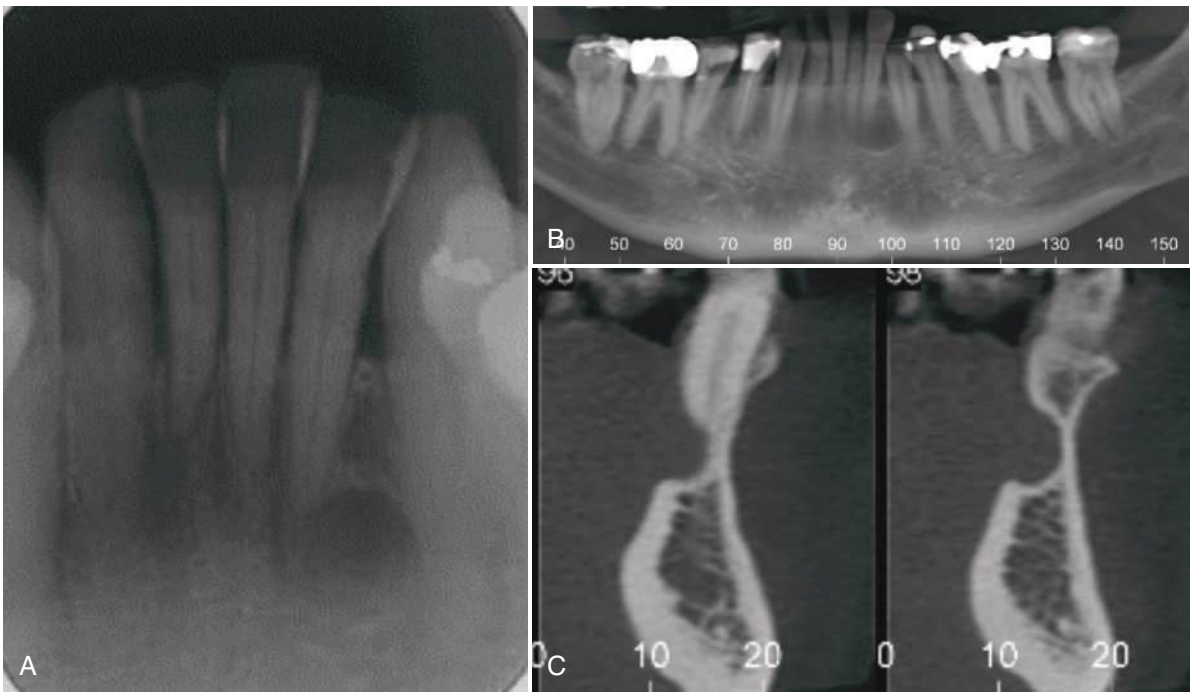


Fig. 3.42 Anterior Stafne defect. **A,** Periapical image exhibiting a well-defined radiolucency near, but not at, the root apex of tooth #23. The lack of restorative history and location of the lesion suggest a noninflammatory etiology and these teeth respond to pulp testing. **B,** Reformatted panoramic image showing a loss of density near the apex of tooth #23. **C,** Coronal CBCT images reveal a concavity in the lingual cortex of bone with no involvement of the root ends. (Courtesy Dr. Bruno Azevedo.)

Summary

For practitioners, it is important to remember that the great majority of radiographic abnormalities encountered on a daily basis will be due to a bacterial etiology as the result of a necrotic pulp. However, it is imperative that clinicians also understand that developmental, metabolic, and neoplastic etiologies can very closely resemble an inflammatory cause.

A multilocular radiolucency should not be the result of a necrotic pulp, and further diagnostic workup is indicated when this type of change is identified. Most radiographic lesions of inflammatory origin are going to be radiolucent, and the presence of calcification suggests a higher likelihood for a nonpulpal etiology. Furthermore, a multifocal presentation should arouse suspicion about noninflammatory etiologies, especially if the dental history and pulp testing are not consistent with pulp necrosis in multiple teeth. Periapical inflammatory disease can be well defined and corticated radiographically or the lesion can have ill-defined margins when the radiographic change is confined to the medullary spaces. However, rapid cortical expansion or perforation of the cortical plate, extensive

tooth mobility, irregular root resorption, and a "moth eaten" appearance suggest a more ominous differential diagnosis.

This chapter categorized lesions based on the most typical radiographic presentations encountered in clinical practice, but many of these pathologies defy classification into a single category and may exhibit a wide variety of radiographic appearances. This review also focused on the most common noninflammatory lesions that present in the periradicular regions, but it is in no way a comprehensive account of all potential pathologies affecting the region.

Radiographs are one piece of the diagnostic puzzle, and accurate interpretation of radiographic findings in conjunction with a thorough review of the medical history, dental history, pulp testing, and clinical examination will yield a differential diagnosis when a nonbacterial etiology is suspected. It is not necessary, nor is it possible, to make a definitive diagnosis of many radiographic abnormalities based exclusively on imaging. However, making the determination that a radiographic lesion is not pulpal in origin should be possible in most cases and further evaluation, in conjunction with referral to the appropriate medical or dental provider, can result in a timely and accurate diagnosis, significantly influencing disease progression and prognosis for many patients.

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